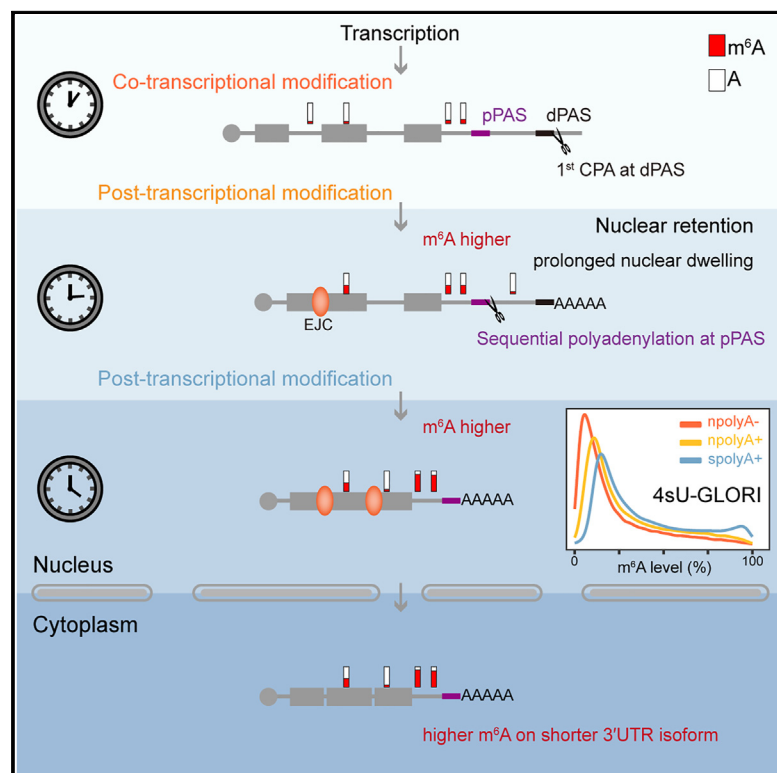


Nuclear retention coupled with sequential polyadenylation dictates post-transcriptional m⁶A modification in the nucleus

Graphical abstract



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In brief

Tang et al. quantify the m⁶A levels for RNA at different stages and reveal widespread post-transcriptional m⁶A modification. They uncover that RNA nuclear retention enhances m⁶A modification and find higher m⁶A levels on shorter 3' UTR isoforms, likely resulting from sequential polyadenylation of longer 3' UTR isoforms with prolonged nuclear dwelling time.

Highlights

- 4sU-GLORI and 4sU-m⁶A-LAIC-seq quantify m⁶A levels of nascent and steady-state RNAs
- Post-transcriptional m⁶A modification is widespread
- RNA nuclear retention enhances m⁶A modification
- Sequential polyadenylation facilitates m⁶A modification

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Article

Nuclear retention coupled with sequential polyadenylation dictates post-transcriptional m⁶A modification in the nucleus

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SUMMARY

N⁶-methyladenosine (m⁶A) modification is deemed to be co-transcriptionally installed on pre-mRNAs, thereby influencing various downstream RNA metabolism events. However, the causal relationship between m⁶A modification and RNA processing is often unclear, resulting in premature or even misleading generalizations on the function of m⁶A modification. Here, we develop 4sU-coupled m⁶A-level and isoform-characterization sequencing (4sU-m⁶A-LAIC-seq) and 4sU-GLORI to quantify the m⁶A levels for both newly synthesized and steady-state RNAs at transcript and single-base-resolution levels, respectively, which enable dissecting the relationship between m⁶A modification and alternative RNA polyadenylation. Unexpectedly, our results show that many m⁶A addition events occur post-transcriptionally, especially on transcripts with high m⁶A levels. Importantly, we find higher m⁶A levels on shorter 3' UTR isoforms, which likely result from sequential polyadenylation of longer 3' UTR isoforms with prolonged nuclear dwelling time. Therefore, m⁶A modification can also take place post-transcriptionally to intimately couple with other key RNA metabolism processes to establish and dynamically regulate epi-transcriptomics in mammalian cells.

INTRODUCTION

N⁶-methyladenosine (m⁶A), the most abundant internal modification in mRNAs, is involved in development,^{1,2} immune activation,³ and cellular response to external stress.⁴ Mechanistically, it can modulate transcription and RNA processing in the nucleus,^{5–11} mRNA nucleus-cytoplasm (CY) shuttling,^{12,13} and mRNA turnover and translation in the CY.^{4,14–19} Transcriptome-wide m⁶A profiling revealed that m⁶A modification selectively occurs in the 3' UTR around stop codons and in long internal exons, on sites with a degenerate consensus sequence DRACH (D = G/A/T, R = A/G, H = A/C/U).^{16,20–22} Interestingly, mRNA m⁶A level is positively correlated with 3' UTR length, leading to its proposed role in repressing polyadenylation in proximal poly(A) site (PAS).²³ However, a follow-up m⁶A profiling on mRNA alternative polyadenylation (APA) isoforms revealed most m⁶A on mRNAs

with shorter 3' UTR, thus suggesting that m⁶A promotes the use of proximal PAS (pPAS).²⁴ To date, there is no experimental evidence to support either paradigm.

The N⁶-Adenosine methylation is catalyzed by m⁶A methyltransferase complex, consisting of the core METTL3-METTL14 heterodimer and other auxiliary factors like WTAP, VIRMA, ZC3H13, etc.^{25–28} The crystal structure of the METTL3-METTL14 complex reveals that METTL3 is the catalytically active subunit, whereas METTL14 functions in maintaining the integrity of the complex and facilitating their binding to the substrate.^{29–31} And through demethylases FTO and ALKBH5, the m⁶A modification can be reversely removed.^{12,32} Accumulating evidence shows that changes in the availability or activity of these writers and erasers would lead to the alternation of m⁶A levels on specific transcripts, thus affecting specific biological processes, such as the cancer progression and embryonic stem cell (ESC)

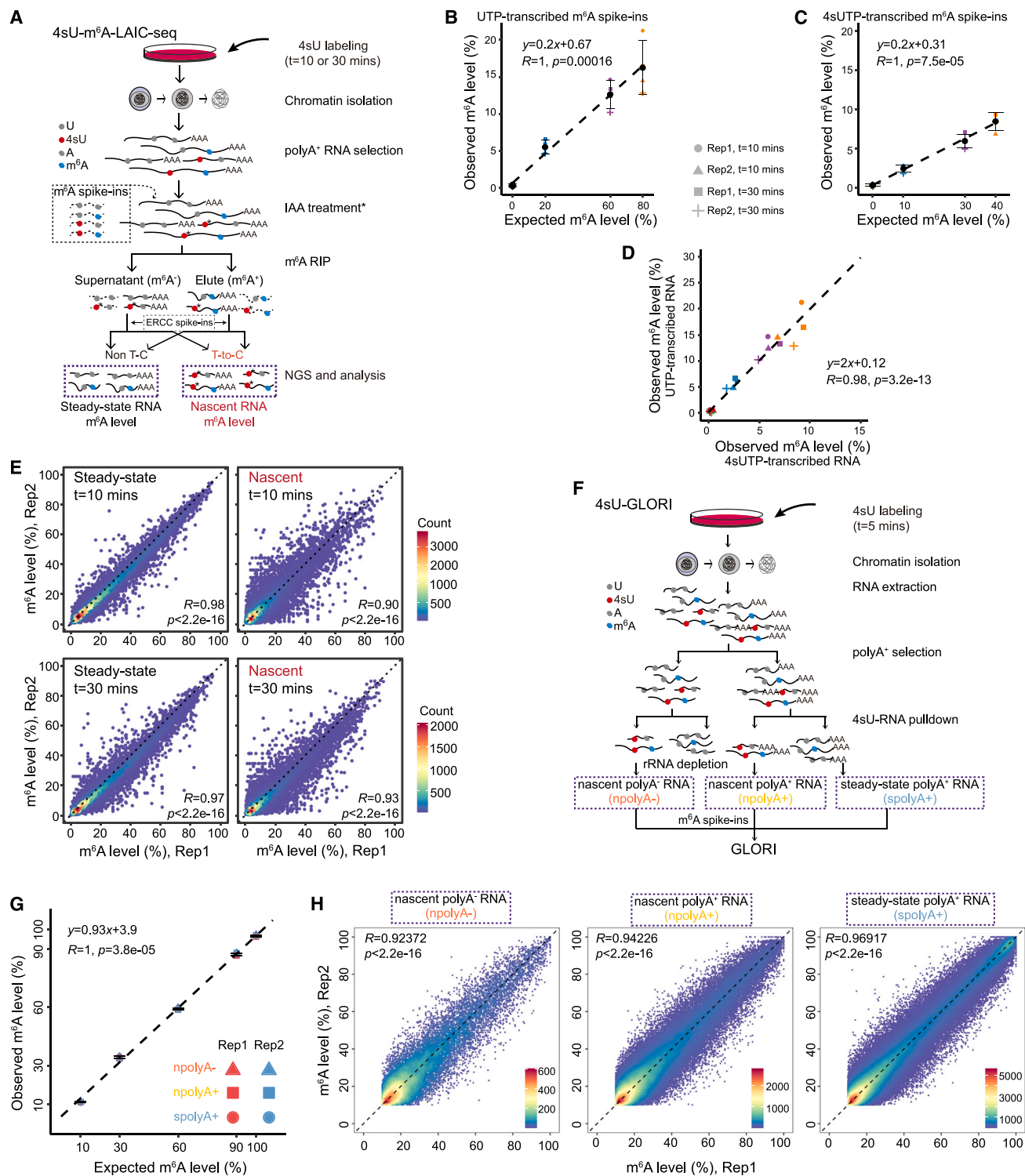


Figure 1. Development of 4sU-m⁶A-LAIC-seq and 4sU-GLORI

(A) Flowchart of 4sU-m⁶A-LAIC-seq. t = 10 or 30 min, cells are incubated with 1 mM 4sU about 10 or 30 min; IAA, iodoacetamide; RIP, RNA immunoprecipitation; m⁶A⁺, m⁶A positive transcripts; m⁶A⁻, m⁶A negative transcripts. Non T-to-C reads and T-to-C reads represent steady-state RNA and newly synthesized RNA, respectively.

(B and C) Scatterplot comparing the expected and observed m⁶A levels of UTP-transcribed m⁶A spike-ins (B) or 4sUTP-transcribed m⁶A spike-ins (C). Rep, biological replicate. Pearson correlation coefficient (R) and p value are indicated, and the error bars represent standard error ($n = 4$).

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pluripotency maintenance.^{33–35} Aside from these mechanisms, transcription factors CEBPZ and SMAD2/3 can facilitate the binding of m⁶A methyltransferase complex to a small subset of genes,^{36,37} while RNA polymerase II (RNA Pol II) and the histone modification H3K36me3 can globally enhance mRNA m⁶A methylation through the recruitment of METTL3 and METTL14, respectively.^{38,39} Collectively, these results indicate that m⁶A modification is co-transcriptionally deposited on pre-mRNAs.

The conundrum might result, at least in part, from mistaking nuclear particle-associated RNAs as chromatin-associated (CA) nascent RNAs. Because the m⁶A level on CA-RNAs is quite similar to that in nucleoplasmic and cytoplasmic RNAs, it is thought that the majority of m⁶A modifications take place co-transcriptionally on nascent RNAs.¹⁵ However, our recent work⁴⁰ and two others^{41,42} revealed that CA-RNAs are not truly equivalent to nascent RNAs; they are actually a mix of nascent transcripts that are still being transcribed by RNA Pol II and processed transcripts that have largely completed splicing and polyadenylation. Therefore, it remains unclear whether m⁶A occurs co- or post-transcriptionally in the nucleus, which is important in understanding the functional interplay between m⁶A modification and RNA processing.

To investigate this problem, we develop 4sU-coupled m⁶A-level and isoform-characterization sequencing (4sU-m⁶A-LAIC-seq) and 4sU-GLORI to quantify the m⁶A levels on steady-state and newly synthesized RNA simultaneously. Our results uncover that for many transcripts/sites, the m⁶A level can be post-transcriptionally enhanced, likely due to prolonged nuclear dwell time, and this process is further linked to the newly elucidated paradigm of sequential polyadenylation by which polyadenylated mRNAs at distal PASs (dPAS) can be further processed at the pPASs before nuclear export.⁴⁰ Our findings shed critical light on the current debate on co- or post-transcriptional m⁶A modification.

RESULTS

Development of 4sU-m⁶A-LAIC-seq

If the majority of m⁶A modification is added to nascent RNAs co-transcriptionally,¹⁵ we would expect that the m⁶A level of newly synthesized polyadenylated RNA is similar to or even higher than steady-state RNA in the chromatin fraction, as m⁶A could accelerate RNA nuclear export or RNA degradation.^{5,12,13} To test this hypothesis, we developed 4sU-m⁶A-LAIC-seq (Figure 1A). In this assay, HeLa cells are cultured in the 4sU (4-thiouridine)-containing medium for 10 or 30 min, followed by isolation of CA polyadenylated RNAs (Figure S1A). These RNAs are then mixed with m⁶A spike-ins and treated with iodoacetamide (IAA) to attach a carboxyamidomethyl group to 4sU, which will induce thymine-to-cytosine (T-to-C) conversion during reverse transcription.⁴³

Last, anti-m⁶A RNA immunoprecipitation (m⁶A RIP) is performed to separate m⁶A modified (m⁶A⁺, elute fraction) and unmodified transcripts (m⁶A[−], supernatant fraction), as described in original m⁶A-LAIC-seq.²⁴ This strategy enables distinguishing newly synthesized RNA (T-to-C reads) from steady-state RNA (non-T-to-C reads) and quantifying the m⁶A levels of nascent RNA vs. steady-state RNA based on the ratio of relative RNA abundances as (elute)/(elute + supernatant).

In our pilot experiments, we treated *in vitro* transcribed RNA with IAA and detected almost no RNA degradation, no matter whether the RNA contained 4sUTP or not (Figure S1B). When RNA was treated with IAA, reverse transcribed, and amplified, the PCR product from 4sUTP-transcribed RNA instead of UTP-transcribed RNA could not be digested with XhoI anymore, indicating that IAA labeling and T-C conversion is highly efficient (Figure S1C), just as reported.⁴³ The m⁶A spike-ins consist of UTP-transcribed RNAs with m⁶A stoichiometry in the range 0%–80% and 4sUTP-transcribed RNA, the level of which is half of their UTP-transcribed counterparts. We computed their m⁶A levels and found that the same spike-in between different samples clustered well both for the UTP-transcribed and 4sUTP-transcribed RNA (Figures 1B and 1C). Although the observed m⁶A level is lower than the actual value (Figures 1B and 1C), possibly due to the inefficient immunoprecipitation of m⁶A antibody, the ratios of m⁶A levels between the UTP-transcribed spike-ins and their 4sUTP-transcribed counterparts are constant as expected (Figure 1D). Thus, 4sU-m⁶A-LAIC-seq can relatively quantify the m⁶A levels of steady-state and newly synthesized RNAs efficiently.

As predicted, the T-to-C conversion ratio is significantly the highest among all twelve nucleotide-nucleotide conversion types (Figure S1D). Interestingly, the ratios of A-to-G and C-to-T types are also higher than the remaining nine other types (Figure S1D), which likely result from A-to-I and C-to-U RNA editing.⁴⁴ In addition, we found that the T-to-C conversion is specific for endogenous 4sU-labeled RNAs and m⁶A spike-ins, as exogenously added ERCC spike-in RNAs exhibit relatively little T-to-C conversion (Figure S1E). Moreover, we examined the distribution of mapped reads and found that there are more intron-derived reads for T-to-C containing reads compared with the non-T-to-C reads (Figure S1F, top: non-T-to-C reads; bottom: T-to-C reads), in line with the presence of more unspliced pre-mRNAs in newly synthesized RNA. Together, these data establish the ability to efficiently discriminate endogenous newly synthesized RNA from steady-state RNA for subsequent m⁶A-LAIC-seq analysis, which shows that m⁶A levels correlated well between biological replicates (Figure 1E; Table S1).

Development of 4sU-GLORI

The m⁶A-antibody-based method is usually confronted with nonspecific binding issues,⁴⁵ and m⁶A-LAIC-seq failed to

(D) Scatterplot comparing the observed m⁶A levels of UTP-transcribed m⁶A spike-ins and 4sUTP-transcribed m⁶A spike-ins.

(E) Scatterplot showing the correlation of m⁶A levels of gene between biological replicates.

(F) Flowchart of 4sU-GLORI. GLORI, glyoxal, and nitrite-mediated deamination of unmethylated adenosines.

(G) Scatterplot comparing the expected and observed m⁶A levels of GLORI m⁶A spike-ins. npolyA[−], nascent polyA-minus RNA; npolyA⁺, nascent polyA-plus RNA; spolyA⁺, steady-state polyA-plus RNA. The error bars represent standard error ($n = 6$).

(H) Scatterplot showing the correlation of m⁶A levels of individual sites between biological replicates for RNA at three different stages.

See also Figures S1 and S2.

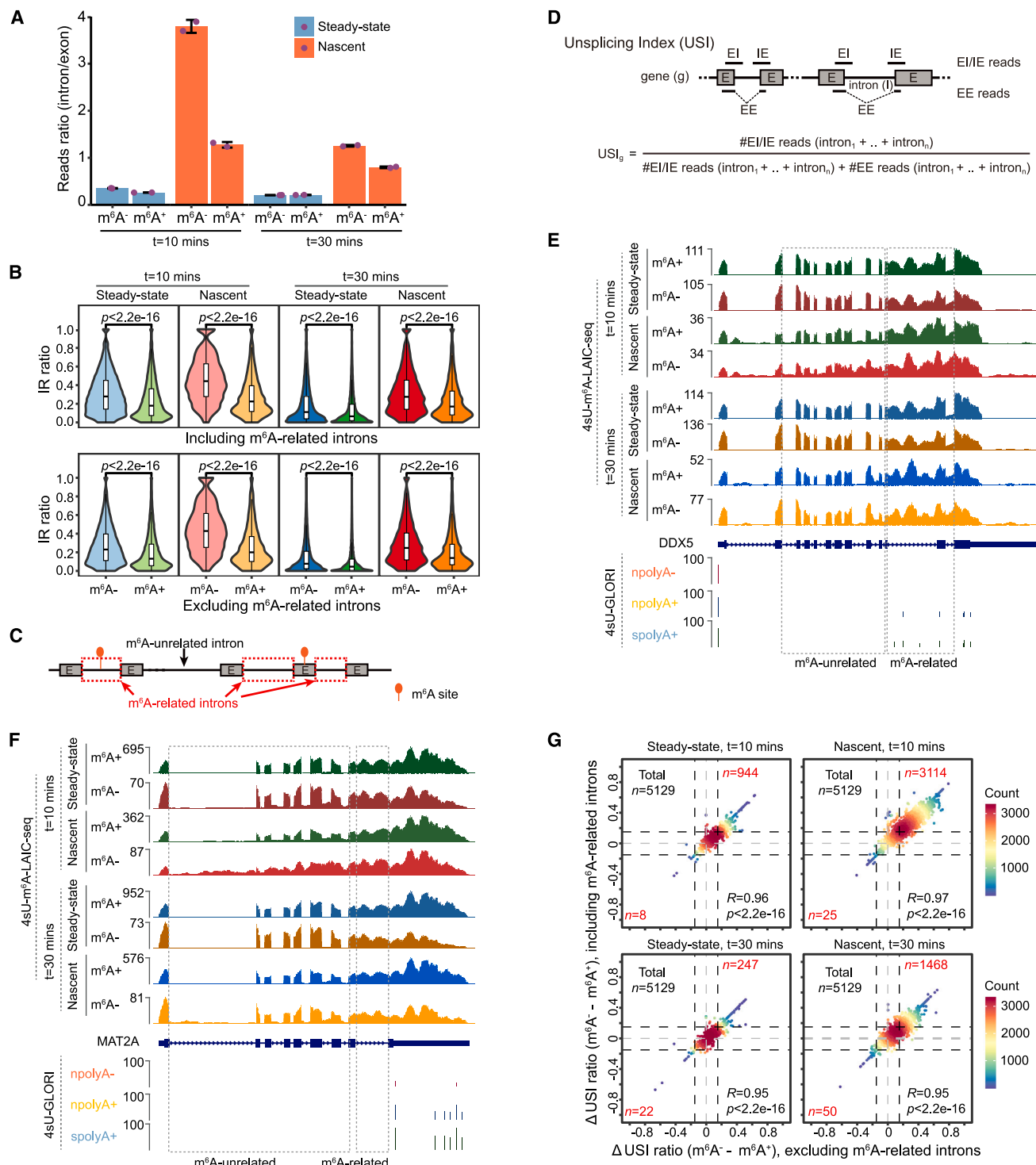


Figure 2. m^6A negative transcripts are more incompletely spliced than m^6A positive transcripts

(A) Bar plot showing the reads ratio of intron relative to exon. Intron reads: exon-intron or intron-exon-junction reads; exon reads: exon-exon reads. m^6A^- , m^6A -negative transcripts; m^6A^+ , m^6A -positive transcripts. t = 10 or 30 min, RNA labeled by 4sU for 10 or 30 min, respectively. The error bars represent standard error.

(B) Violin plot showing the intron retention (IR) ratio of individual introns between m^6A^- and m^6A^+ transcripts for steady-state RNA and nascent RNA. The IR ratio represents the extent of intron retention, and the m^6A -related introns refer to the introns with detected m^6A sites within the intron itself or in the adjacent exons. The p value is calculated based on Wilcoxon signed-rank tests.

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quantify m⁶A levels at single-nucleotide resolution. Thus, we developed 4sU-GLORI to quantify the m⁶A levels of newly synthesized and steady-state RNA at single-nucleotide resolution based on GLORI (glyoxal and nitrite-mediated deamination of unmethylated adenosines)⁴⁶ (Figure 1F). We first validated the specificity of 4sU-RNA pull-down (Figures S1G and S1H) and the efficiency of 4sU labeling (Figures S1I and S1J). In the 4sU-GLORI assay, HeLa cells are cultured in the 4sU-containing medium for 5 minutes, followed by isolation of CA RNAs, which are further separated into nascent polyA-minus RNA (npolyA⁻), nascent polyA-plus RNA (npolyA⁺), and steady-state polyA-plus RNA (spolyA⁺) based on polyA selection and 4sU RNA pull-down.⁴⁷ Then, these three RNA populations are subject to RNA sequencing (RNA-seq) and the GLORI.⁴⁶

As exemplified by six genes in Figure S2A, npolyA⁻ RNA displayed significant signal downstream transcription termination site (TTS), while npolyA⁺ and spolyA⁺ RNAs showed almost no signal downstream TTS, consistent with the nascent and mature features of polyA⁻ and polyA⁺ RNAs, further confirmed by a global analysis of the RNA-seq signals around the PAS sites (Figure S2B). In line with these, we found that the fraction of reads supporting cleavage and polyadenylation is significantly higher in npolyA⁺ and spolyA⁺ samples compared with the npolyA⁻ sample (Figure S2C). Moreover, the majority of introns from npolyA⁻ and spolyA⁺ RNAs are almost un-spliced and spliced, respectively, while the introns of npolyA⁺ are partially spliced, and global analysis also shows the same characteristics (Figures S2D and S2E). Together, these results showed the expected features and confirmed the authenticity of npolyA⁻, npolyA⁺, and spolyA⁺ RNAs.

We computed the m⁶A levels of spike-in RNAs and found excellent agreement between the expected and actual values (Figure 1G), reflecting that GLORI can quantify the absolute levels of m⁶A sites.⁴⁶ We identified 16,750, 70,175, and 137,212 m⁶A sites, which are shared by biological replicates for npolyA⁻, npolyA⁺, and spolyA⁺ RNAs, respectively (Figure S2F; Table S2), and their levels correlated well between biological replicates (Figure 1H; Table S2). Further analysis revealed that these m⁶A sites exhibit typical features of m⁶A sites with consensus sequence DRACH (D = G/A/T, R = A/G, H = A/C/U, Figure S2G) and are enriched around stop codon (Figure S2H), reflecting that they are authentic m⁶A sites.

m⁶A-negative transcripts are more incompletely spliced

Anticipated from co-transcriptional m⁶A modification, we were surprised to find a higher proportion of exon-intron (EI) or intron-exon (IE) junction reads in m⁶A-negative transcripts for newly synthesized RNA, especially for 10 min-labeled RNA (Figures 2A and S1F, T-to-C reads), suggesting that m⁶A-nega-

tive transcripts are likely to be more incompletely spliced than m⁶A-positive transcripts. Biological replicates correlated well (Figure 1E), and we combined them for further analysis. We calculated the intron retention (IR) ratio of individual introns using IRFinder,⁴⁸ finding that the IR ratios from the m⁶A-negative transcripts are generally higher, no matter whether the m⁶A-related introns were considered or not (Figures 2B and 2C). We further analyzed the splicing completeness for individual genes using the unsplicing index defined in Figure 2D. As shown for genes *DDX5* and *MAT2A* (Figures 2E and 2F), m⁶A-negative transcripts are more incompletely spliced, especially among 10 min-labeled nascent RNAs, which reflects a global trend (Figure 2G, top; Table S3).

There are two potential mechanistic explanations for the aforementioned observation. m⁶A can accelerate intron removal, just as reported,⁴⁹ or alternatively, m⁶A modification takes place post-transcriptionally for many m⁶A sites. If the former was true, we would expect to see introns adjacent to m⁶A sites (m⁶A-related) but not the others (m⁶A-unrelated) to be regulated (Figures 2C, 2E, and 2F). In other words, the extent of splicing for m⁶A-unrelated introns should have no difference for polyA⁺ RNA if m⁶A is largely co-transcriptionally installed. Nevertheless, after excluding the introns with detected m⁶A sites within the intron or in the adjacent exons by utilizing the identified m⁶A sites from our GLORI data (Figure 2C), we still observed the same trend in individual genes (10 min-labeled nascent RNA, Figures 2E and 2F) or in global analysis (Figures 2B and 2G). These observations thus point to the possibility that m⁶A may be largely deposited on spliced and polyadenylated mRNAs after transcription at least for some genes.

Widespread post-transcriptional m⁶A modification

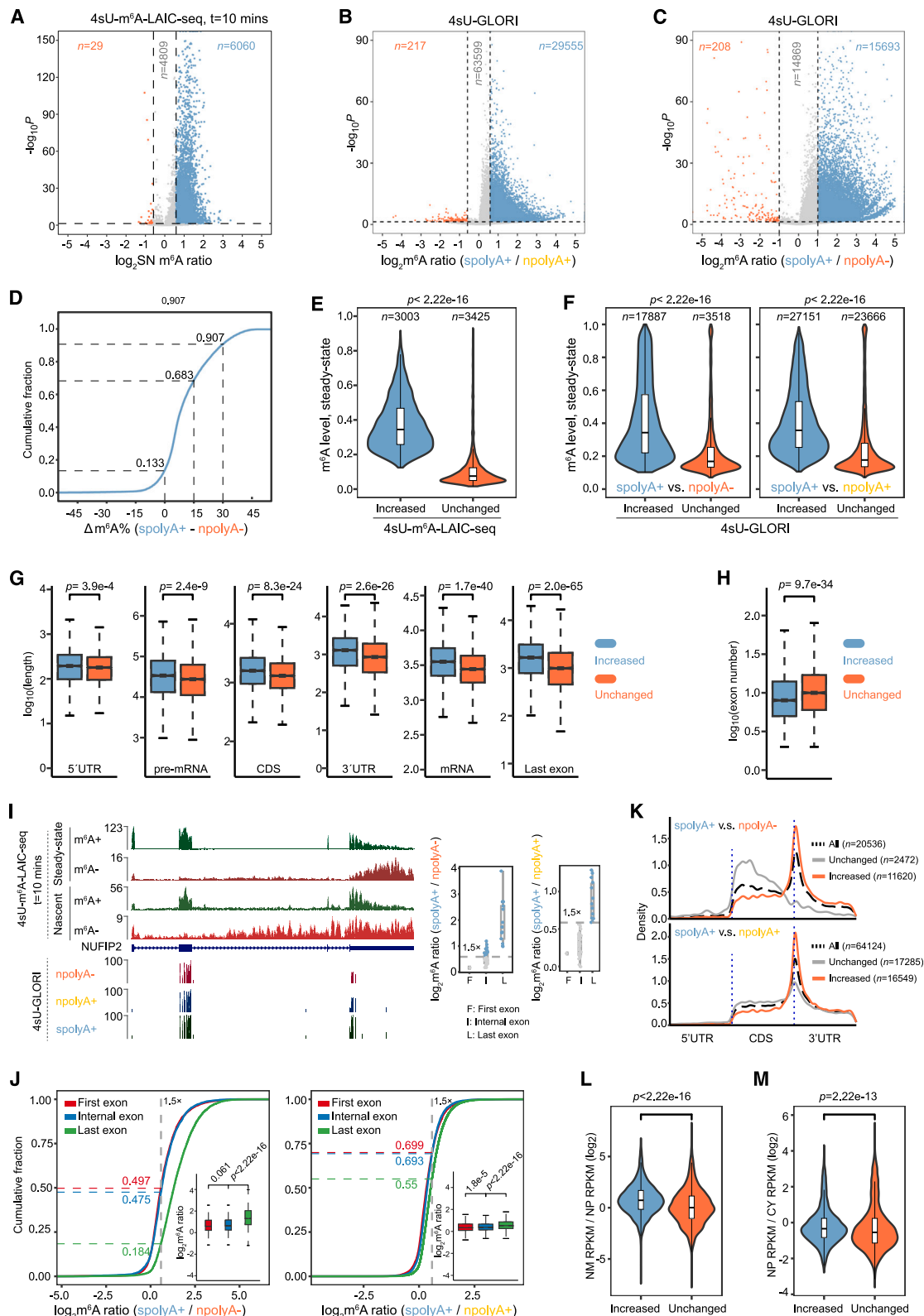
To directly evaluate the extent of post-transcriptional m⁶A modification, we calculated the SN m⁶A ratio (m⁶A level of steady-state RNA over that of newly synthesized RNA), a value that reflects the extent of post-transcriptional m⁶A modification. We observed that 6,060 (~55.6%) protein-coding genes showed a higher SN m⁶A ratio within 10 min of 4sU-labeling under the cutoff of 1.5-fold, whereas only 29 genes exhibited the converse pattern (Figure 3A; Table S1). This suggests that RNAs for most of the detected genes are further modified by m⁶A after transcription. Consistent with this, we observed the T-to-C conversion ratio (indicative of the extent of nascent RNA over steady-state RNA) of m⁶A-negative transcripts is about 2.38-fold higher than the m⁶A-positive transcripts (Figure S1E), meaning that nascent RNAs are less m⁶A modified. We observed the same but not so significant trend in samples with 30 min of 4sU-labeling (Figures S1E, S3A, and S3B; Table S1), which is expected as more occurrence of post-transcriptional m⁶A modification during

(C) Schematic representation of m⁶A-related introns (in dotted line rectangles) and m⁶A-unrelated introns. The box and line represent exon and intron, respectively.

(D) Definition of the gene unsplicing index (USI) metric. The EI/IE and EE reads represent exon-intron (EI)/intron-exon (IE) junction reads and exon-exon (EE) junction reads, respectively.

(E and F) Representative gene examples showing normalized RNA-seq signals from 4sU-m⁶A-LAIC-seq for *DDX5* (E) and *MAT2A* (F). m⁶A sites identified from 4sU-GLORI are shown below, and the height represents the relative m⁶A level.

(G) Scatterplot showing the correlation of USI difference between m⁶A-negative and m⁶A-positive transcript calculated with/without the m⁶A-related introns. The gray and black dashed lines represent the difference values 0 and 0.15/−0.15, respectively. Pearson correlation coefficient (*R*) and *p* values are indicated.



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longer 4sU incubation. In a given fixed 4sU incubation time, the SN m^6A ratio of longer transcripts is expected to be larger than shorter ones if m^6A addition can occur post-transcriptionally (Figure S3C). This is largely what we observed (Figure S3D).

For m^6A level comparison from 4sU-GLORI data, we combined all m^6A sites, identified at least in one of the three samples, and then kept those with coverage higher than 30 in all compared samples for further analysis. We observed a gradual increase of m^6A level from npolyA⁻ to npolyA⁺ and from npolyA⁺ to spolyA⁺ RNA (Figures S3E and S3F). Compared with npolyA⁺ RNA, 31.6% m^6A sites exhibit 1.5-fold higher m^6A -level in spolyA⁺ RNA (higher m^6A -level sites), while the opposite is true for 0.2% m^6A sites (Figure 3B; Table S2). All the sites used for comparison belong to 8,578 genes, among which 6,211 genes (~72%) own higher m^6A -level sites (Figure S3G). Thus, both 4sU-GLORI and 4sU- m^6A -LAIC-seq data demonstrated that many transcripts undergo post-transcriptional m^6A modification.

Next, we compared the npolyA⁻ and spolyA⁺ RNA GLORI data, which is more suitable for quantifying the extent of post-transcriptional m^6A modification. 51% m^6A sites exhibit 2-fold higher m^6A -level in spolyA⁺ RNA, while the opposite is true for 0.7% m^6A sites (Figure 3C; Table S2), indicating that m^6A methylation occurs more after transcription for half of the sites. Moreover, even for the sites below that threshold, the level of 32% sites in the spolyA⁺ RNA increased by at least 15% compared with npolyA⁻ RNA (Figure 3D). All the sites used for comparison belong to 4,348 genes, among which 3,348 (~77%) genes own higher m^6A -level sites (Figure S3H). Additionally, we inspected the gene *MAT2A* (Figure 2F), the m^6A sites of which are catalyzed by METTL16,¹⁰ and found that the methylation of all detected m^6A sites largely occurs post-transcriptionally (Figure S3I). Together, we can conclude that the kinetics of m^6A modification is not so fast as suspected, and post-transcriptional m^6A modification is widespread.

Features of post-transcriptional m^6A modification

We next sought to characterize the features associated with post-transcriptional m^6A modification. We first defined m^6A increased transcripts/sites (m^6A ratio ≥ 1.5 and m^6A difference is more than 10%) and m^6A unchanged genes/sites (m^6A ra-

tio < 1.5 and m^6A difference is between -5% and 5%). We found that the m^6A level of steady-state RNA is higher for m^6A -increased transcripts/sites compared with m^6A -unchanged transcripts/sites (Figures 3E and 3F). Gene Ontology (GO) analysis revealed that m^6A high genes (the genes of 3,000 most methylated transcripts based on m^6A level at steady state) and m^6A -increased genes almost share the same enriched terms, while m^6A -low genes (the genes of 3,000 least methylated transcripts) and m^6A -unchanged genes almost share the same enriched terms (Figure S4A). These data together suggest that transcripts/sites with high m^6A levels are more likely to undergo post-transcriptional m^6A modification.

As m^6A modification is positively correlated to 3' UTR length while negatively correlating to exon numbers,²²⁻²⁴ we examined these features and found that m^6A -increased transcripts generally exhibit longer 3' UTR and fewer exons than m^6A -unchanged genes, and last exon length showed the strongest correlation (Figures 3G and 3H; Table S1). We performed the same analysis on transcripts with/without increased m^6A sites (Figures S3G and S3H), and the results confirmed the strongest positive correlation of last exon length for post-transcriptional m^6A modification (Figures S4B and S4C). Thus, it seems that genes with longer terminal exons are more likely to be modified post-transcriptionally.

Individual inspection of the gene *NUFIP2* shows that m^6A sites, located at the last exon, are more prone to undergo post-transcriptional m^6A modification compared with those at first or internal exons (Figure 3I), which is confirmed by global analysis (Figure 3J). Density plots of the increased and unchanged m^6A sites showed that increased m^6A sites are more enriched at the 3' UTR, also suggesting that m^6A sites at the last exon are more likely to be modified post-transcriptionally (Figure 3K). Furthermore, it seems that post-transcriptional m^6A modification is independent of the position (Figure S4D) or the exact motif for m^6A sites in the last exons (Figure S4E).

Transcripts with high m^6A levels are more restricted in the nucleus

Next, we attempted to understand the mechanism responsible for post-transcriptional m^6A modification. Given that m^6A writers

Figure 3. Characterization of the genes or sites prone to be modified by m^6A post-transcriptionally

- (A) Volcano plot comparing the m^6A level of steady-state and nascent RNA. $t = 10$ min, RNA labeled by 4sU for 10 min; SN m^6A ratio, m^6A level ratio of steady-state RNA relative to nascent RNA. p value is calculated by Fisher's exact test and adjusted based on the Benjamin-Hochberg model.
- (B and C) Volcano plot comparing the m^6A levels of individual sites for RNA at different stages: spolyA⁺ vs. npolyA⁺ (B) and spolyA⁺ vs. npolyA⁻ (C).
- (D) Cumulative curve showing the m^6A difference between spolyA⁺ and npolyA⁻ RNA for those sites below the threshold (fold-change of SN m^6A ratio < 2 , orange and gray dots in C).
- (E) Violin plot comparing the m^6A levels of genes at steady-state for m^6A increased and unchanged genes. Increased transcripts, SN ratio > 1.5 and $|m^6A$ level difference| > 0.1 ; unchanged transcripts, SN ratio < 1.5 and $|m^6A$ level difference| < 0.05 . p value: Wilcoxon signed-rank tests.
- (F) Violin plot comparing the m^6A levels of individual sites at steady-state for m^6A increased and unchanged sites. Increased sites, m^6A ratio > 1.5 and $|m^6A$ level difference| > 0.1 ; unchanged sites, m^6A ratio < 1.5 and $|m^6A$ level difference| < 0.05 .
- (G and H) Boxplot comparing gene features between m^6A -increased and -unchanged genes. Boxplot definition is the same as Figure S3. p values: Wilcoxon signed-rank tests.
- (I) UCSC genome browser views showing 4sU- m^6A -LAIC-seq and 4sU-GLORI signals for RNAs at different stages and boxplots comparing the m^6A ratios of individual sites located at different exons for gene *NUFIP2*.
- (J) Cumulative curve comparing the m^6A ratios of individual sites located at different exons.
- (K) Density plot showing the distribution of all, unchanged, and increased m^6A sites.
- (L and M) Violin plot comparing the relative enrichments of polyA⁺ RNA in different cellular fractions (NM vs. NP in L; NP vs. CY in M) between m^6A -increased and -unchanged genes. CY, cytoplasm; NP, nucleoplasm; NM, nuclear matrix.
- See also Figures S3 and S4.

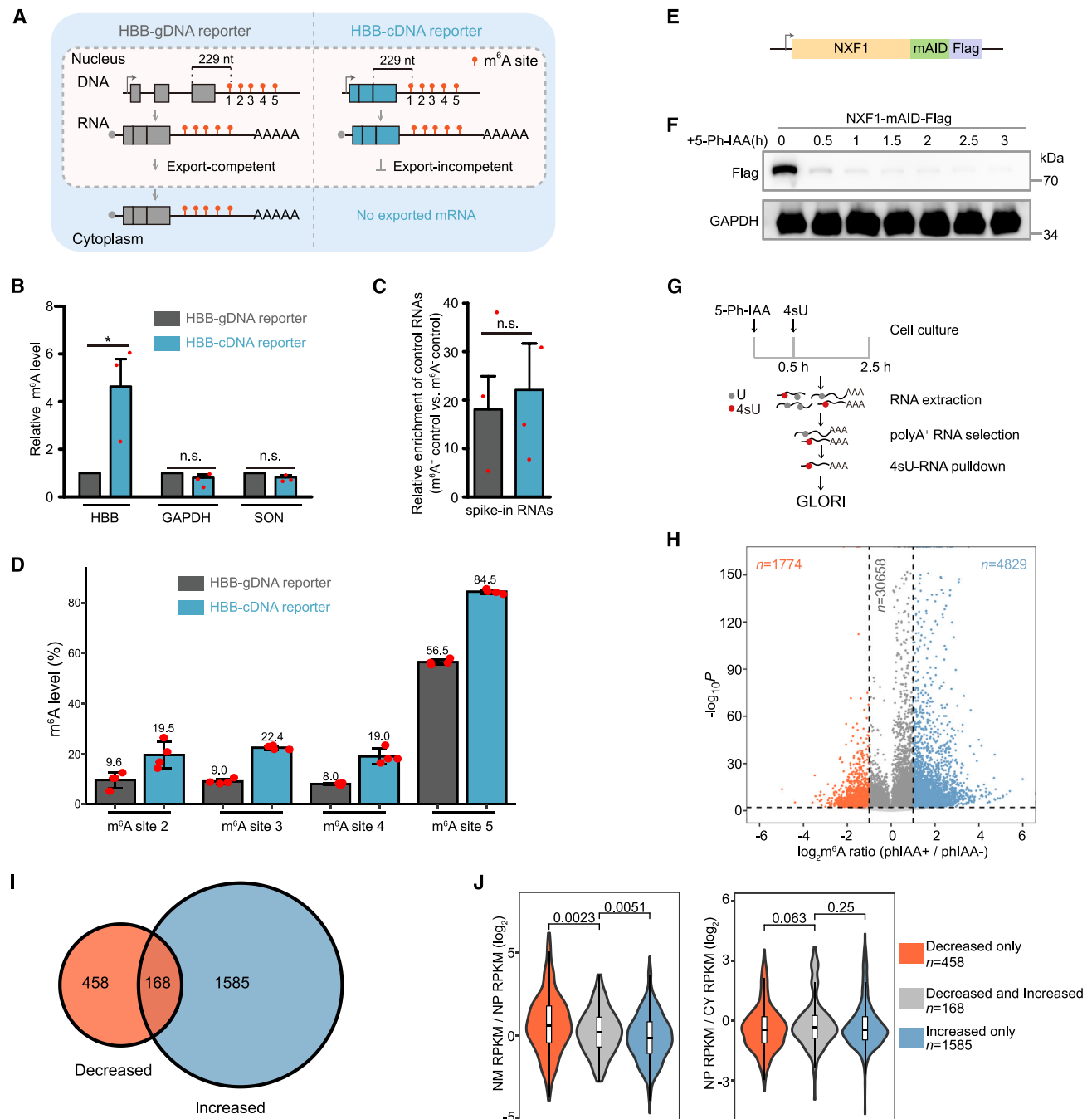


Figure 4. RNA nuclear retention enhances m⁶A modification

(A) Schematic representation of intron-containing HBB-gDNA and intronless HBB-cDNA reporters.

(B) Relative m⁶A levels of the HBB RNAs and two endogenous RNAs (GAPDH and SON) from the cells transfected with the HBB-gDNA or HBB-cDNA reporters. The normalized m⁶A levels of RNAs in the cells transfected with the HBB-gDNA were arbitrarily set to 1.

(C) RT-qPCR results showing the enrichment of exogenous m⁶A positive control RNA relative to the exogenous m⁶A negative control RNA. The exogenous control RNAs were added before IP in (B). Data are presented as mean values $\pm$ SD ($n = 3$). p values: a two-tailed unpaired t test. * p value < 0.05 ; n.s., not significant.

(D) The m⁶A levels of four sites for HBB reporters quantified by region-specific GLORI.

(E) Diagram of NXF1-mAID-FLAG cell line. mAID, mini AID, for inducing the degradation of target protein upon the binding of 5-Ph-IAA.

(F) Western blot showing the depletion efficiency of NXF1 at different times after 5-Ph-IAA addition.

(G) Flowchart of GLORI with/without NXF1 depletion.

(H) Volcano plot comparing the m⁶A levels of individual sites with/without NXF1 depletion. phIAA+, 5-Ph-IAA phIAA-, control.

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are enriched in nuclear speckles,^{25–27} we reasoned that if a transcript is quickly transported to the CY once completing transcription, there would be little chance for post-transcriptional m⁶A modification. Previously, we separated HeLa cells into the CY, nucleoplasm (NP), chromatin, and nuclear matrix (NM) and found there was almost no RNA uniquely associated with the chromatin fraction.⁴⁰ In fact, CA RNA analyzed in most studies is all tightly associated with NM.⁴⁰ Utilizing these data, we calculated the relative RNA enrichment in NP (NP/CY ratio) and NM (NM/NP ratio) and found that m⁶A-increased transcripts identified from 4sU-m⁶A-LAIC-seq are more significantly positively correlated with the NM/NP ratio compared with the NP/CY ratio (Figures 3L and 3M; Table S4). These observations indicate a positive link between post-transcriptional m⁶A modification and the time of RNA residing in chromatin and/or NM, consistent with the previous finding that the m⁶A methylation is essentially completed before RNAs are released into the NP.¹⁵

To further investigate the putative role of nuclear dwell time in enhancing m⁶A modification, we compared the m⁶A levels of mRNAs and long intergenic noncoding RNAs (lincRNAs), the latter of which are usually more enriched in the nucleus⁵⁰ (Figure S5A). Results show that the m⁶A level of lincRNA is generally higher than mRNA (Figure S5B). The upstream antisense RNAs (uaRNAs), which are often restricted in the nucleus by the Mtr4/ZFC3H1 complex,⁵¹ also exhibit higher m⁶A levels than their protein-coding counterparts (Figures S5C and S5D).

As splicing facilitates mRNA export,⁵² which implies that intronless genes may be less efficiently exported or require a specialized pathway(s) for export. Given the m⁶A reader YTHDC1 can assist export of m⁶A modified transcripts,¹³ we sought to assess the impact of YTHDC1 on exporting intronless vs. intron-containing genes. Upon YTHDC1 knockdown, intronless genes are more impacted than intron-containing genes, even for those in group 10, the average m⁶A level of which is higher than intronless genes (Figures S5E and S5F). Furthermore, metagene analysis revealed that the m⁶A level of intronless genes is higher than intron-containing genes (Figure S5G). Therefore, we propose that intronless genes are more dependent on m⁶A modification and YTHDC1 for efficient export. Collectively, these data reveal a potential dynamic interrelationship between post-transcriptional m⁶A modification, RNA nuclear retention, and export.

RNA nuclear retention enhances m⁶A modification

Exon-junction complex (EJC), deposited on spliced RNA following intron removal, can suppress m⁶A methylation,^{53–55} which may also contribute to the observed higher m⁶A levels for intronless protein-coding genes or intron-poor lincRNA/uaRNA. To determine the causal relationship, we constructed a pair of human β -globin (HBB) chimeric reporters containing five m⁶A sites from *c-myc*,⁵⁶ all of which are more than 200-nt far away from the start site of the last exon. In these constructs, the HBB-gDNA reporter contains its natural introns, whereas the

HBB-cDNA reporter lacks the introns (Figure 4A). Although both constructs generate identical mature RNA, only the RNA generated from the HBB-gDNA reporter can be efficiently exported into the CY.⁵²

We performed m⁶A-RIP followed by quantitative reverse transcription PCR (RT-qPCR) and found that the m⁶A level of export-incompetent RNA (HBB-cDNA reporter) is more than 4-fold higher than the export-competent RNA (HBB-gDNA reporter) (Figure 4B). This effect was not due to different immunoprecipitation efficiency between samples, as both the endogenous gene GAPDH and SON (Figure 4B) and the exogenously m⁶A positive control showed almost the same immunoprecipitation efficiency (Figure 4C). Furthermore, we could quantify the m⁶A levels of four sites by region-specific GLORI and found that the levels of all these four sites are higher for export-incompetent RNA (Figure 4D), indicating that RNA nuclear retention can enhance m⁶A modification.

Next, we applied auxin-inducible degron 2 (AID2) technology for the acute depletion of NXF1, the essential mRNA export factor.^{57,58} We generated NXF1-mAID-FLAG cell line, in which NXF1 was fused with miniAID degron (mAID) and FLAG tag in frame (Figure 4E). Cells were treated with/without 5-Ph-IAA for 0.5 h, which led to almost complete depletion of NXF1 (Figure 4F), then 4sU was added and cultured for another 2 h, and after that, nascent polyA⁺ RNA was purified and subject to GLORI (Figure 4G). We found that the m⁶A levels of 4,829 and 1,774 sites are increased or decreased at least 2-fold, respectively, upon NXF1 depletion (Figure 4H; Table S2). Those increased or decreased sites are from 1,753 and 626 genes, respectively (Figure 4I). Moreover, the genes with increased sites have a lower NM/NP ratio but no difference in NP/CY ratio compared with those with decreased sites (Figure 4J), suggesting that transcripts of these genes, quickly released from NM to NP, may have little chance for post-transcriptional modification, and blocking RNA export would provide more such opportunity and enhance their m⁶A levels. Thus, both individual reporter assay and global analysis from NXF1 depletion prove that RNA nuclear retention can enhance m⁶A modification.

m⁶A modification has little impact on APA

During this investigation, we observed that the shorter 3' UTR isoforms of some genes are more m⁶A-modified than their longer counterparts (m⁶A⁺ shorter genes, exemplified by gene *TRAF6* in Figure 5A), and the converse is true for others, in which the longer 3' UTR isoforms are more m⁶A-modified than their shorter counterparts (m⁶A⁺ longer genes, exemplified by gene *NANS* in Figure 5B). To systematically investigate these differences, we computed the percentage of distal isoform usage index (PDUI) values for both m⁶A⁺ and m⁶A[−] transcripts and required the PDUI difference ≥ 0.15 . We identified 1,463 m⁶A⁺ shorter genes and 629 m⁶A⁺ longer genes for 10 min-labeled samples (Figure 5C; Table S5), and the numbers are 1,556 and 561, respectively, for 30 min-labeled samples (Figure 5D; Table S5). Thus,

(I) The gene numbers with m⁶A-level-altered sites upon NXF1 depletion. Increased: blue dots in (H); Decreased: orange dots in (H).

(J) Violin plot comparing the relative enrichments of polyA⁺ RNA in different cellular fractions for genes in different groups categorized in (I). CY, cytoplasm; NP, nucleoplasm; NM, nuclear matrix. *p* values: Wilcoxon signed rank tests.

See also Figure S5.

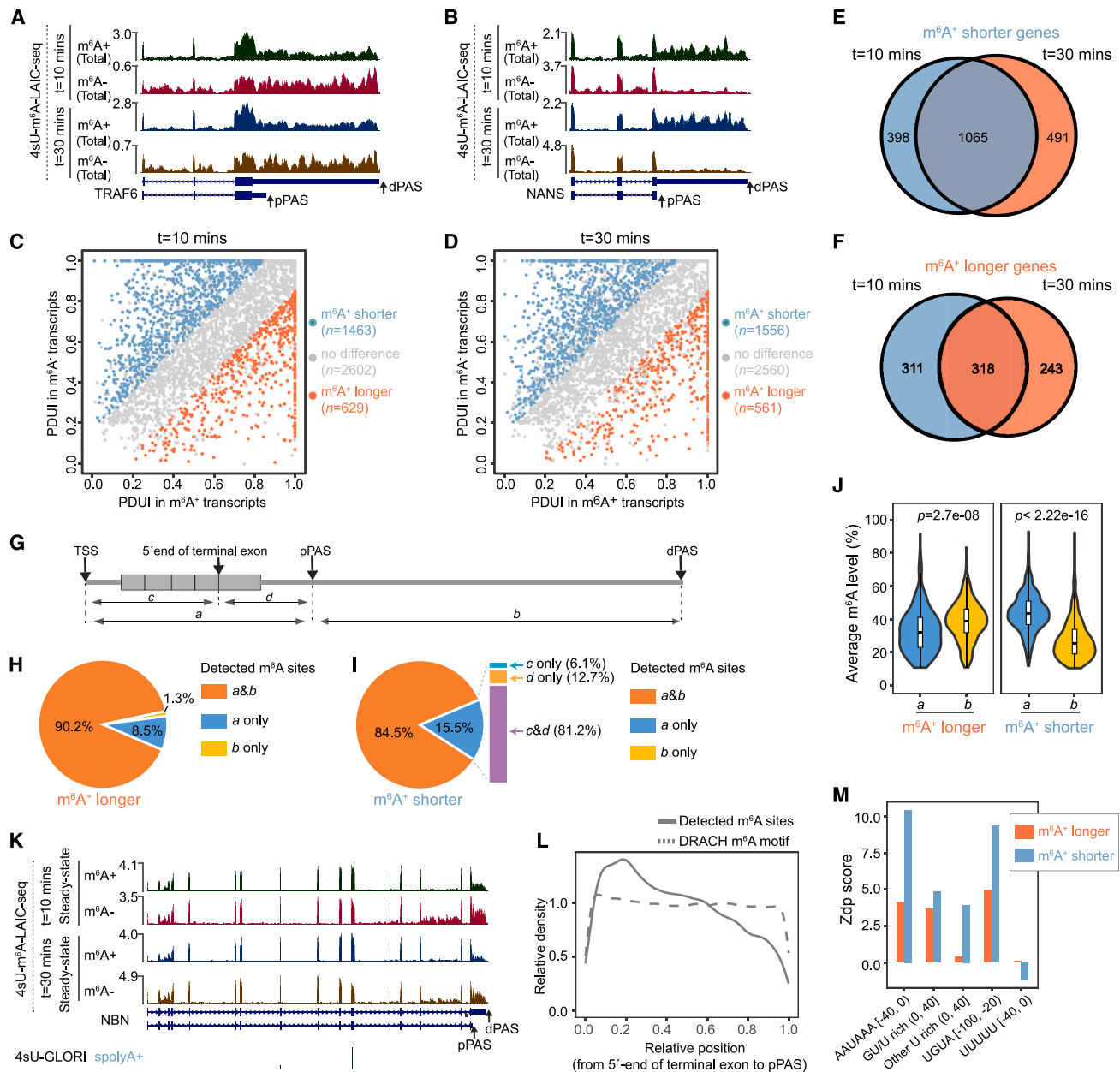


Figure 5. Relationship between m^6A modification and alternative polyadenylation

(A and B) UCSC genome browser views of *TRAF6* (A) and *NANS* (B) showing the RNA-seq signals of m^6A^+ and m^6A^- transcripts around the proximal PAS (pPAS) and distal PAS (dPAS).

(C and D) Global comparison of the PDI values of m^6A^+ and m^6A^- transcripts. $t = 10$ or 30 min, RNA labeled by 4sU for 10 or 30 min, respectively. The orange and blue dots represent the genes with a PDI difference ≥ 0.15 and false discovery rate (FDR) ≤ 0.05 using Fisher's exact test with Benjamini-Hochberg correction.

(E and F) Venn diagram showing the overlapped m^6A^+ shorter genes (E) and m^6A^+ longer genes (F) identified in (C) and (D).

(G) Schematic representation of the four divided regions between the transcript start site (TSS), 5' end of the terminal exon, pPAS, and dPAS.

(H and I) Pie charts showing the proportion of genes in different classes according to the locations of their exonic m^6A sites for the m^6A^+ longer genes (H) and for the m^6A^+ shorter genes (I).

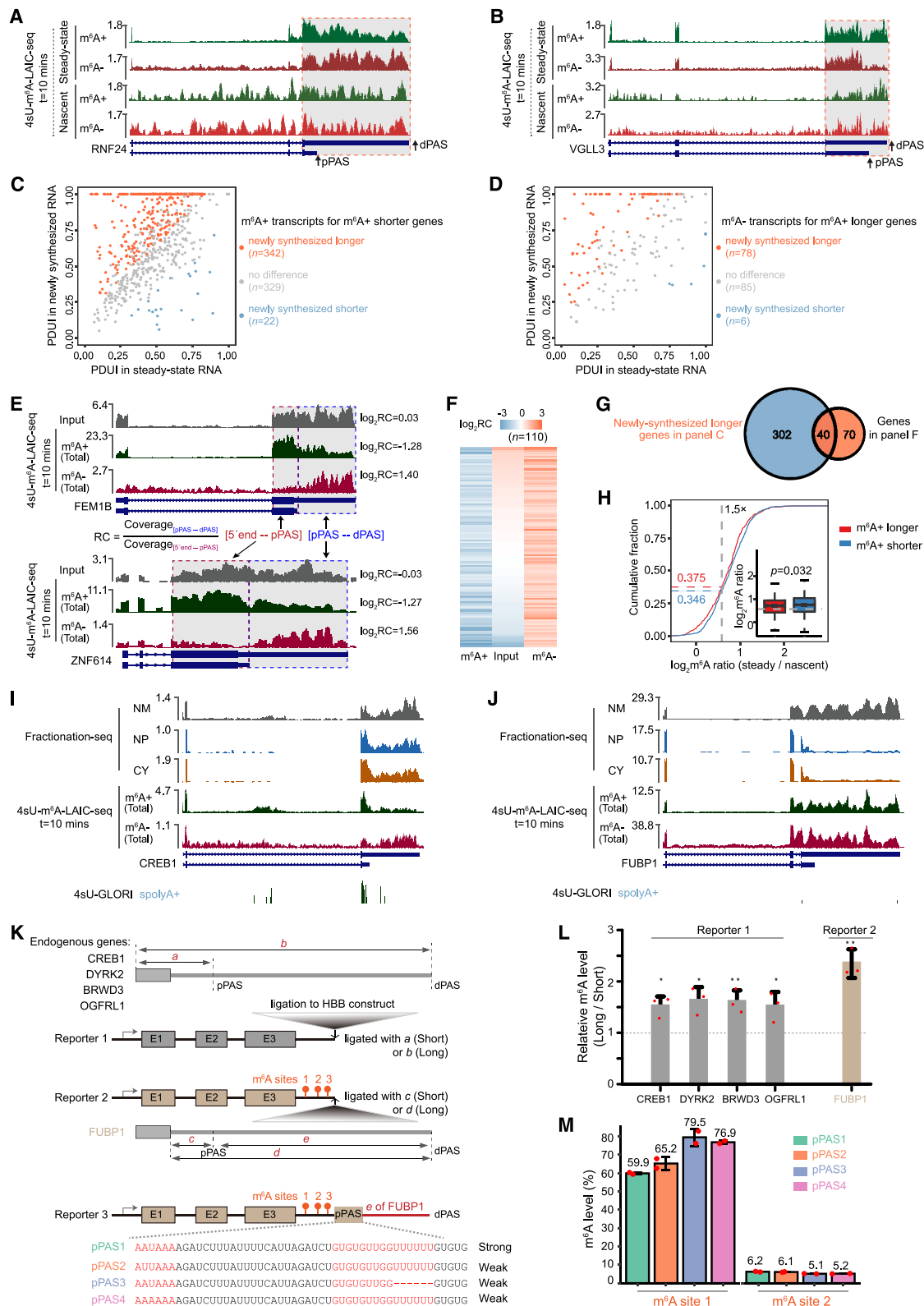
(J) Violin plot comparing the average levels of m^6A sites located at regions *a* and *b* for m^6A^+ shorter and longer genes. p values: Wilcoxon signed rank tests.

(K) UCSC genome browser views showing the detected exonic m^6A sites far away from polyadenylation sites for the m^6A^+ shorter genes *NBN*.

(L) Relative position of the detected m^6A sites and the DRACH motif within the region *d* (from the 5' end of the terminal exon to pPAS).

(M) Z_{dp} values between dPAS and pPAS of different motifs for m^6A^+ longer genes (orange) and m^6A^+ shorter genes (blue).

See also Figure S6.



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transcripts with shorter 3' UTR are preferentially modified by m⁶A, as reported.²⁴ Among them, 1,065 genes are m⁶A⁺ shorter genes (Figure 5E), while 318 genes are m⁶A⁺ longer genes (Figure 5F) for both samples, which are used for downstream analysis.

We divided all transcripts into two regions (region *a* from transcript start site [TSS] to pPAS and region *b* from pPAS to dPAS) (Figure 5G) and detected m⁶A sites at both regions for the majority of m⁶A⁺ shorter and longer genes (Figures 5H and 5I). However, we found that the average level of the m⁶A sites at region *b* is higher than those at region *a* for m⁶A⁺ longer genes (Figure 5J, left), which may contribute to the observation that their longer 3' UTR isoform is preferentially modified. By contrast, the converse is true for m⁶A⁺ shorter genes (Figure 5J, right). Given the entire sequences in shorter 3' UTR isoforms are shared with the longer ones, we hypothesized that m⁶A modification upstream of the pPAS might promote its utilization, such that the unmodified transcript will use the dPAS. To test this hypothesis, we knocked down METTL3 (Figure S6A) and observed a decreased m⁶A level quantified by liquid chromatography-tandem mass spectrometry (LC-MS/MS) (Figure S6B). Then we performed RNA-seq on RNAs extracted from different fractionations. Results showed the APA pattern was altered only in a limited number of genes, either for m⁶A⁺ shorter or longer genes (Figures S6C and S6D). Thus, it seems that m⁶A does not selectively promote the usage of pPAS.

We then considered where certain APA factors assembled around the pPAS may facilitate m⁶A modification through cross-talk with the m⁶A methyltransferase complex. In such a case, it is supposed that DRACH motifs adjacent to pPAS are more m⁶A-modified. We divided region *a* into region *c* (from the TSS to the beginning of the last exon) and region *d* (from the beginning of the last exon to the pPAS), as depicted in Figure 5G. We found that a small fraction of m⁶A⁺ shorter genes, whose m⁶A sites can only be detected at region *c* (Figure 5I), as exemplified by gene *NBN* (Figure 5K). Meanwhile, we found a progressive decrease in the m⁶A distribution toward the pPASs within region *d*, while the m⁶A motif DRACH is evenly distributed (Figure 5L). Together, this evidence does not support that APA factors assembled around the pPAS facilitate m⁶A modification.

Sequential polyadenylation generates m⁶A-modified shorter 3' UTR isoforms

Next, we asked whether APA might impact m⁶A. We found that in general, the pPAS is less frequently used than the dPAS among m⁶A⁺ shorter genes compared with m⁶A⁺ longer genes (Figure S6E). Consistent with this observation, the canonical polyadenylation signal (AAUAAA/AUUA) is also more prevalently associated with dPASs over pPASs (Figure S6F). We further used Z_{dp} value to quantify the difference of polyadenylation enhancing elements between dPAS and pPAS,⁵⁹ finding that the dPAS is generally more associated with these positive regulatory elements for both m⁶A⁺ shorter and m⁶A⁺ longer genes, but the difference is larger for the m⁶A⁺ shorter genes (Figure 5M).

Given our recent discovery of sequential APA, where the dPAS is used first in a default reaction and, through nuclear retention, the distally processed mRNA is further polyadenylated at the pPAS,⁴⁰ we hypothesized that this post-transcriptional mechanism might contribute to prevalent m⁶A modification on mRNAs with shorter 3' UTRs. As shown for gene *RNF24*, although its shorter 3' UTR isoform is more m⁶A-modified at steady state (Figure 6A, steady state), there is no such difference for newly synthesized RNA (Figure 6A, nascent). In gene *VGL33* where m⁶A is more associated with longer 3' UTR isoform at steady state (Figure 6B, steady state), we also detected no such difference in newly synthesized RNA (Figure 6B, nascent). Through global analysis of the genes with sufficient coverage in newly synthesized RNA (mean coverage ≥ 5), we found that 342 out of 693 (49.4%) m⁶A⁺ shorter genes for the m⁶A⁺ transcripts (Figure 6C; Table S5) and 78 out of 169 (46.2%) m⁶A⁺ longer genes for the m⁶A⁺ transcripts (Figure 6D; Table S5) display more dPAS usage in the newly synthesized RNA population ($PDU_{\text{newly synthesized}} - PDU_{\text{steady-state}} \geq 0.15$). These data again suggest that the pPAS should be sequentially activated in APA without being impacted by m⁶A.

For some m⁶A⁺ shorter genes, we noted that the mean coverage of the region from the beginning of the terminal exon to the pPAS (upstream region) is almost the same as that in the

Figure 6. Sequential polyadenylation facilitates m⁶A modification

(A and B) UCSC genome browser views showing the m⁶A⁺ and m⁶A⁻ RNA-seq signals for both the nascent and steady-state RNA in gene *RNF24* (A) and *VGL33* (B).

(C and D) Global comparison of the PDUI values in the newly synthesized and steady-state RNA of the m⁶A⁺ transcripts for the m⁶A⁺ shorter genes (C) and of the m⁶A⁻ transcripts for the m⁶A⁺ longer genes (D). The m⁶A⁺ longer or shorter genes represent the genes whose longer or shorter 3' UTR isoforms are preferentially more m⁶A modified, respectively. The orange and blue dots represent the genes with a PDUI difference ≥ 0.15 and FDR ≤ 0.05 using the Fisher's exact test with Benjamini-Hochberg correction.

(E) UCSC genome browser views showing the complementary coverage pattern from the 5' end of the terminal exon to dPAS in the m⁶A⁺ and m⁶A⁻ samples relative to the input for genes *FEM1B* (top) and *ZNF614* (bottom). The input sample refers to the sample prior to m⁶A-RIP. RC, relative coverage metric.

(F) Heatmap of $\log_2(\text{RC})$ values for 110 genes with the complementary coverage pattern as in (E).

(G) Venn diagram showing the overlap of the identified genes in (C) and (F).

(H) Comparison of the m⁶A levels of newly synthesized RNA and steady-state RNA for the m⁶A⁺ shorter (left) and m⁶A⁺ longer genes (right). *p* value: Wilcoxon signed-rank test.

(I and J) UCSC genome browser views showing the cellular distribution of longer and shorter 3' UTRs isoforms and the methylation states of longer and shorter 3' UTRs isoforms in the chromatin for the gene *CREB1* (I) and gene *FUBP1* (J). The detected m⁶A sites are shown at the bottom.

(K) Schematic representation of different reporters. Refer to STAR Methods for details of reporters.

(L) Relative m⁶A level of the cytoplasmic RNA from the cells integrated with the different reporters. The normalized m⁶A levels of RNA from the short reporters were arbitrarily set to 1. Data are presented as mean values $\pm$ SD (*n* = 3). *p* values: a two-tailed unpaired *t* test. * *p* < 0.05; ** *p* < 0.01.

(M) The m⁶A levels of two sites for HBB reporter 3 quantified by region-specific GLORI. The error bars represent standard error.

See also Figure S7.

region from the pPAS to the dPAS (downstream region) prior to m⁶A RIP (input), but the coverage of the upstream region became much higher than the downstream region in the m⁶A-selected sample, and the converse was true in the m⁶A flowthrough sample, as shown on gene *FEM1B* and *ZNF614* (Figure 6E), which strongly indicate the existence of 3' cleaved fragments as a result of sequential polyadenylation. We calculated the relative coverage (RC) between the downstream region to the upstream region and required $RC \leq 2/3$ in the m⁶A-selected sample, $RC \geq 1.5$ in the m⁶A flowthrough sample, and the RC of the input sample within the range of the two RCs. There are 110 genes satisfying those criteria (Figure 6F), of which 40 were processed at the dPASs in the newly synthesized RNA population (Figure 6G). Together, these strongly support that those m⁶A-modified mRNAs ended at pPASs are derived from sequential polyadenylation.

Sequential polyadenylation facilitates m⁶A modification

To provide further evidence for the contribution of sequential polyadenylation to m⁶A modification, we examined the cellular distribution of transcripts for m⁶A⁺ shorter genes and found that at least 174 genes whose longer 3' UTR isoforms were selectively enriched in the NM (Figures S7A–S7C), especially in the nuclear speckles (Figure S7D). Additionally, we observed that the SN m⁶A ratios for m⁶A⁺ shorter genes are larger than m⁶A⁺ longer genes (Figure 6H). To directly demonstrate the role of sequential APA in regulating m⁶A levels, we randomly selected four m⁶A⁺ shorter genes from the 174 candidate genes, *CREB1* (Figure 6I), *DYRK2*, *BRWD3*, and *OGFRL* (Figure S7E). We inserted their terminal exons of short or long APA isoforms in the 3' UTR of an HBB lentivirus reporter (Figure 6K, reporter 1). We also selected another low m⁶A level gene, *FUBP1* (Figure 6J), which undergoes sequential APA,⁴⁰ to insert its short or long 3' UTR into the HBB lentivirus reporter that contains validated m⁶A sites (Figure 6K, reporter 2).

We generated multiple cell lines with these reporters integrated into the genome, extracted their cytoplasmic RNA, and mixed them prior to RIP as diagrammed in Figure S7F. Results of m⁶A RIP-RT-qPCR showed that the m⁶A level of RNA arising from the long 3' UTR reporter is usually 1.5- to 2-fold higher than that from the short counterpart (Figure 6L).

Additionally, we inserted a strong pPAS,⁶⁰ which is supported to interrupt sequential polyadenylation, together with the extended UTR of *FUBP1* into the HBB lentivirus reporter (Figure 6K, pPAS1 of reporter 3), meanwhile we weakened this pPAS by mutation (AAUAAA to AUUAAA, pPAS2) or U-rich element deletion (pPAS3),^{60–62} which enables the occurrence of sequential polyadenylation. Alternatively, we abolish its processing activity by another mutation (AAUAAA to AAAAAA, pPAS4).⁶² We could quantify the m⁶A level of two sites by region-specific GLORI and found that the level of the site with high m⁶A level can increase by 20% (pPAS3 vs. pPAS1), similar to pPAS4, while the site with low m⁶A level is not increased (Figure 6M), suggesting that the level of sites with high m⁶A level can be enhanced through sequential polyadenylation. We conclude that mRNA ended at the dPAS facilitates m⁶A modification, which is further processed into the shorter isoforms ended at the pPAS to be exported into the CY.

DISCUSSION

m⁶A modification has been implicated in various physiological activities.^{2–4,63} However, many questions still remain with respect to when and where m⁶A is deposited on pre-mRNAs. In this study, we developed 4sU-m⁶A-LAIC-seq and found that the m⁶A level of spolyA⁺ is at least 1.5-fold higher than newly synthesized polyA⁺ RNA for 6,060 (55.6%) detected protein-coding genes in HeLa cells, suggesting m⁶A modification still occurs post-transcriptionally. This point was independently corroborated by 4sU-GLORI, which quantified the levels of single m⁶A sites for npolyA[–], npolyA⁺, and spolyA⁺ RNA. Furthermore, it seems that m⁶A methylation largely occurs post-transcriptionally for 51% of the detected m⁶A sites (polyA[–] vs. spolyA⁺ RNA). These results are in contrast to the popular model of co-transcriptional m⁶A modification.¹⁵

Similar to post-transcriptional splicing and polyadenylation,^{40,64,65} post-transcriptional m⁶A modification may offer versatile strategies to regulate m⁶A levels, such as RNA nuclear retention demonstrated here. This is a way quite different from the co-transcriptional associated mechanisms.^{36–39} Similar to post-transcriptional splicing,⁶⁵ we suspect that post-transcriptional m⁶A modification may also take place in nuclear speckles where all components of the m⁶A methyltransferase complex are concentrated.^{25–27}

In agreement with post-transcriptional m⁶A deposition, m⁶A negative transcripts are more incompletely spliced than the m⁶A positive transcripts for polyadenylated RNAs. Photoactivatable-ribonucleoside enhanced crosslinking and immunoprecipitation (PAR-CLIP) was used to capture the targets of WTAP and METTL3, revealing that their major binding sites are in exonic regions after normalizing exon and intron length.²⁶ Additionally, methylated RIP (MeRIP) performed on CA RNAs^{11,15} or bromouridine (BrU)-enriched nascent RNA⁴⁹ also showed that exon is greatly favored for m⁶A addition over intron. Thus, it appears that for many genes, m⁶A deposition follows the removal of most introns except those flanking post-transcriptionally spliced exons.^{64,65} Thus, exon is selectively methylated, and the intronic m⁶A is usually associated with alternatively spliced regions.^{11,26,49} Furthermore, accompanied by intron excision, EJC is deposited on spliced transcripts, which can coat relatively long mRNA stretches,⁶⁶ thereby shielding potential m⁶A sites in most short exons,^{53–55} consistent with the observation that long internal exons rather than short ones appear more prone to be m⁶A-modified.^{20,22} These mechanisms may collectively contribute to selective m⁶A modification in 3' UTRs.

m⁶A modification can regulate splicing,^{9,10,67,68} such as the last intron excision of *MAT2A*.¹⁰ However, another investigation revealed that m⁶A only plays a limited role in splicing.¹⁵ Thus, the results are conflicted about the relationship between m⁶A modification and alternative splicing. Here, we found that the methylation of all detected m⁶A sites for gene *MAT2A* mostly occurs post-transcriptionally (Figures 2F and S3I), thereby excluding the possibility that m⁶A regulates the splicing of *MAT2A* co-transcriptionally. Instead, as shown in Figure 2F, all introns except the last are almost spliced in m⁶A positive transcripts, indicating that m⁶A modification

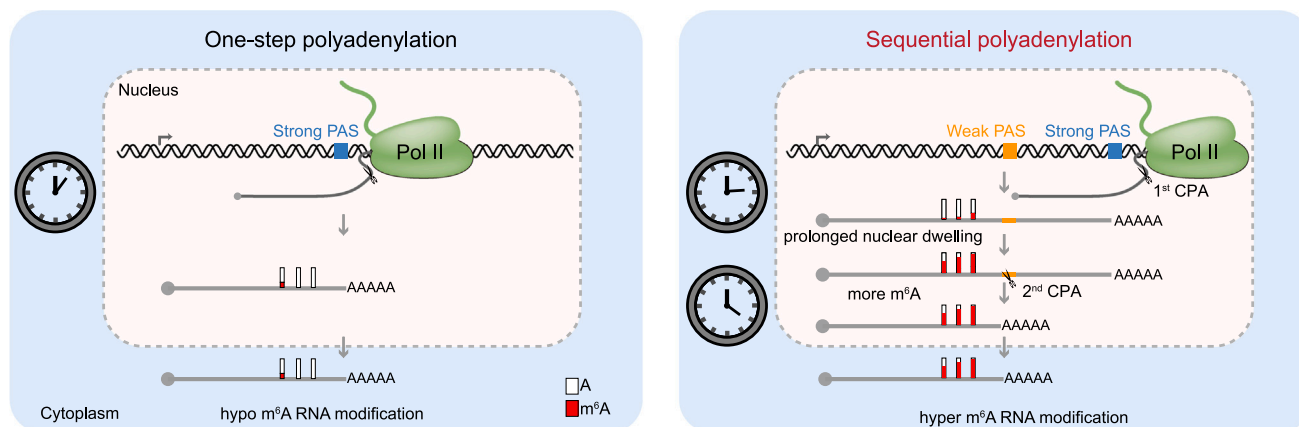


Figure 7. Model of sequential polyadenylation in facilitating m^6A modification

Sequential polyadenylation (right) is proposed to endow the RNA with more nuclear dwelling time to be m^6A -modified post-transcriptionally compared with a one-step polyadenylation process (left).

precedes ahead of last intron excision and then promotes its removal post-transcriptionally. Together, it is more likely that m^6A is restricted to regulate a subset of post-transcriptional splicing events.

m^6A level is positively correlated to 3' UTR length, and METTL3 knockdown would lead to altered APA pattern for 111 genes, in which 70 favored a more usage of pPAS.²³ These results supported the role of m^6A modification in repressing proximal polyadenylation. However, another study revealed shorter 3' UTR isoforms are preferentially modified by m^6A for hundreds of genes,²⁴ which argues against a general role of m^6A modification in repressing proximal polyadenylation but hints that m^6A modification may facilitate the usage of pPAS. However, the authors only found a limited number of genes with altered APA patterns after analyzing m^6A writer knock-down or knockout datasets.²⁴ Hence, the authors suspected that APA factors may promote m^6A modification,²⁴ but this paradigm lacks experimental support and could not account for the positive relationship between m^6A level and 3' UTR length.^{23,24}

Our data confirm that shorter 3' UTR isoforms are preferentially modified, and m^6A plays a limited role in APA.²⁴ For m^6A^+ shorter genes, m^6A sites showed a pattern enriched around 5' end of the last exon instead of pPAS, refutes the hypothesis that the APA factors promote m^6A modification through crosstalk with the m^6A methyltransferase complex or vice versa.²⁴ We observed that transcripts with longer 3' UTR are usually retained in the NM-associated nuclear speckles and are more prone to undergo post-transcriptional m^6A modification. Meanwhile, a pairwise comparison of the relationship between m^6A modification and RNA isoforms for newly synthesized vs. steady-state RNA revealed that m^6A modification does not influence APA in most cases, but the converse is true that initial longer mRNA isoforms ended at the dPAS may be retained in the nucleus for post-transcriptional m^6A modification, and subsequent polyadenylation helps release the shorter transcripts for export (Figure 7). We further established this model through HBB reporters, and our results

showed that the m^6A levels of transcripts derived from reporters with weak rather than strong pPAS are higher, directly supporting the role of sequential polyadenylation in facilitating m^6A modification and further denying that APA factors promote m^6A modification. We speculate that a similar mechanism may also apply to other RNA modification events.⁴⁴ Our mechanistic study thus provides a new framework to understand the functional interplay of m^6A and other RNA modifications with other RNA metabolism pathways under physiological or pathological conditions.

Limitations of the study

We found that some factors may cause the extent of post-transcriptional modification to be underestimated, such as a relatively long labeling time used in 4sU- m^6A -LAIC-seq, insufficient discrimination of nascent RNA and steady-state RNA due to the relatively low 4sU incorporation rate, and potential mRNA nuclear degradation induced by m^6A modification.⁵

To elucidate the role of sequential polyadenylation in facilitating m^6A modification, polyA⁺ RNAs were preferred in 4sU- m^6A -LAIC-seq. However, polyA[−] RNA and spolyA⁺ should be the better choice for quantifying the extent of post-transcriptional m^6A modification. 4sU-GLORI may compensate for this problem, but the potential contamination of steady-state RNA in the purified nascent RNA may compromise the result.⁶⁹ On the other hand, the sequenced reads are relatively short (the median length is about 40 nt) in GLORI.⁴⁶ Consequently, we could not identify and compare the methylation levels of all m^6A sites in the transcriptome due to a limited sequencing depth. For the same reason, we miss a few m^6A sites in our reporters, as it is difficult to specifically amplify the products from the reporters without the interference of endogenous fragments with the same sequence.

Although our model proposes that sequential polyadenylation coupled with nuclear retention dictates post-transcriptional m^6A modification, which contributes to the observation that m^6A is positively associated with the length of last exon, while the shorter 3' UTR isoforms are preferentially methylated,^{23,24} we

do not exclude the possibility that m⁶A might regulate APA or vice versa under some circumstances.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.molcel.2024.07.017>.

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AUTHOR CONTRIBUTIONS

P.T. and Y.Z. conceived the study. P.T. performed most of the experiments with the help of H.Z., L.H., F.L., B.D., and G.L. J.Y., Z.C., C.D., Y.Y., and Y.Z. analyzed the sequencing data. T.S. and X.B. contributed critical experimental information. P.T. and Y.Z. wrote the manuscript with inputs from Z.C. and J.Y. All authors discussed and approved the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Rabbit anti-METTL3	Proteintech	Cat# 15073-1-AP; RRID: AB_2142033
Rabbit anti-GAPDH	Proteintech	Cat# 10494-1-AP; RRID: AB_2263076
Mouse anti-Flag	Sigma	Cat# F1804; RRID: AB_262044
Rabbit anti-m ⁶ A	Synaptic Systems	Cat# 202003; RRID: AB_2279214
Mouse anti-m ⁶ A	New England Biolabs	Cat# E1610S; RRID: AB_2923416
Mouse anti-SC35	Sigma	Cat# S4045; RRID: AB_477511
Anti-Digoxin	Invitrogen	Cat# 700772; RRID: AB_2532342
Bacterial and virus strains		
PLVX-HBB-CREB1-short lentivirus	This paper	N/A
PLVX-HBB-CREB1-long lentivirus	This paper	N/A
PLVX-HBB-DYRK2-short lentivirus	This paper	N/A
PLVX-HBB-DYRK2-long lentivirus	This paper	N/A
PLVX-HBB-BRWD3-short lentivirus	This paper	N/A
PLVX-HBB-BRWD3-long lentivirus	This paper	N/A
PLVX-HBB-OGFRL1-short lentivirus	This paper	N/A
PLVX-HBB-OGFRL1-long lentivirus	This paper	N/A
PLVX-HBB-ppp1r8-FUBP1-short lentivirus	This paper	N/A
PLVX-HBB-ppp1r8-FUBP1-long lentivirus	This paper	N/A
PLVX-HBB-ppp1r8-PAS1-FUBP1-long lentivirus	This paper	N/A
PLVX-HBB-ppp1r8-PAS2-FUBP1-long lentivirus	This paper	N/A
PLVX-HBB-ppp1r8-PAS3-FUBP1-long lentivirus	This paper	N/A
PLVX-HBB-ppp1r8-PAS4-FUBP1-long lentivirus	This paper	N/A
Chemicals, peptides, and recombinant proteins		
BglII	Thermo Fisher	Cat# FD0084
Sall	Thermo Fisher	Cat# FD0644
AgeI	Thermo Fisher	Cat# FD1464
Lipofectamine™ 2000	Thermo Fisher	Cat# 11668019
Lipofectamine™ RNAiMAX	Thermo Fisher	Cat# 13778150
Puromycin	MedChemExpress	Cat# HY-B1743A
TRIzol Reagent	Thermo Fisher	Cat# 15596018
Trizol LS	Thermo Fisher	Cat#10296028CN
ImProm-II™ Reverse Transcriptase	Promega	Cat# A3803
Taq DNA Polymerase	Thermo Fisher	Cat# EP0402
Nuclease P1	New England Biolabs	Cat# M0660S
Alkaline Phosphatase	Takara	Cat# 2250A
HEPES	Sigma	Cat# 54457
NP-40	Biosharp	Cat# BS205
KCl	SINOPHARM	Cat# 10016318
EDTA	Thermo Fisher Scientific	Cat# AM9260G
DTT	Sigma	Cat# 43816

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Sucrose	Amresco	Cat# M117
Tris-HCl	Thermo Fisher Scientific	Cat# 15567027
NaCl	Sigma	Cat# 71386
cOmplete™ EDTA-free protease inhibitor cocktail	Roche	Cat# 04693132001
Glycerol	Biosharp	Cat# BS154
MgCl ₂	Thermo Fisher	Cat# AM9530G
Urea	Sigma	Cat# U5378
Turbo DNase I	Thermo Fisher	Cat# AM2239
T7 RNA Polymerase	New England Biolabs	Cat# M0251S
Ribonucleotide Solution Set	New England Biolabs	Cat# N0450S
4sUTP	Jena Bioscience	Cat# NU-1156S
m ⁶ ATP	Synthgene	Cat# m ⁶ ATP001-1
Ethanol	SINOPHARM	Cat# 10009164
Iodoacetamide	Sigma	Cat# I1149
NaH ₂ PO ₄	Sigma	Cat# S3139
Na ₂ HPO ₄	Sigma	Cat# S0876
DMSO	Sigma	Cat# D8418
BSA	Amresco	Cat# 9048-46-8
SSPE	Sigma	Cat# S2015
Tween-20	Biosharp	Cat# BS100
SDS	Thermo Fisher	Cat# AM9820
Phenol: chloroform: isopropyl alcohol	Solarbio life sciences	Cat# P1014
Glycogen	Thermo Fisher	Cat# R0551
H ₂ O	Thermo Fisher	Cat#10977023
ERCC RNA Spike-In Control Mixes	Invitrogen	Cat# 4456740
MTSEA Biotin-XX	UElandy	Cat# M5037
Thermostable RNase H	Beyotime	Cat# R7090S
Glyoxal	Sigma	Cat# 50649
H ₃ BO ₃	Sigma	Cat# B0394
NaNO ₂	Sigma	Cat# 31443
MES	Sigma	Cat# V900336
TEA	Sigma	Cat# 90335
Formamide (Deionized)	Thermo Fisher	Cat# AM9342
T4 Polynucleotide Kinase	New England Biolabs	Cat# M0201S
SUPERase.in	Thermo Fisher	Cat# AM2696
T4 RNA Ligase 2, truncated KQ	New England Biolabs	Cat# M0373L
5' deadenylase	New England Biolabs	Cat# M0331S
RecJf	New England Biolabs	Cat# M0264S
ProtoScript® II Reverse Transcriptase	New England Biolabs	Cat# M0368L
dNTP Set Solution	Yeaston	Cat# 10122ES74
T4 RNA Ligase 1 (ssRNA Ligase), High Concentration	New England Biolabs	Cat# M0437M
Lithium chloride solution	Sigma	Cat# L7026
RNasin Ribonuclease inhibitor	Promega	Cat# N2515
Critical commercial assays		
pEASY-basic seamless cloning and assembly kit	TransGen Biotech	Cat# CU201
2× SYBR Green qPCR Mix	Yeaston	Cat# 11201ES03
RNA Clean and Concentrator 5	Zymo research	Cat# R1014

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
NEBNext® Ultra™ II directional RNA library prep with sample purification beads	New England Biolabs	Cat# E7765S
Oligo d(T)25 Magnetic Beads	New England Biolabs	Cat# S1419S
Protein A/G magnetic beads	Thermo Fisher	Cat# 88802
MaXtract High Density	Qiagen	Cat# 129046
Agencourt RNA Clean XP beads	Beckman Coulter	Cat# A63987
RNA HS Assay Kit	Thermo Fisher	Cat# Q32852
Dynabeads® Antibody Coupling Kit	Thermo Fisher	Cat# 14311D
Dynabeads™ MyOne™ Streptavidin C1 beads	Thermo Fisher	Cat# 65002
RNA fragmentation reagents	Thermo Fisher	Cat# AM8740
oligo Clean and Concentrator	Zymo research	Cat# D4061
ExoSAP-IT	Thermo Fisher	Cat# 78200

Deposited data

Raw sequencing data	Tang et al. ⁴⁰ ; Roundtree et al. ¹³ ; This paper	GEO: GSE165742; GEO: GSE74397; GSA-human: HRA006964
Raw western and microscopy images	This paper	Mendeley Data, https://doi.org/10.17632/x5njxrdpm5.2

Experimental models: Cell lines

Human HeLa	NCACC	Cat# TCHu187
Human HEK293T	NCACC	Cat# GNHu44
Human HeLa NXF1-AID2	This paper	N/A
Human HeLa PLVX-HBB-CREB1-short	This paper	N/A
Human HeLa PLVX-HBB-CREB1-long	This paper	N/A
Human HeLa PLVX-HBB-DYRK2-short	This paper	N/A
Human HeLa PLVX-HBB-DYRK2-long	This paper	N/A
Human HeLa PLVX-HBB-BRWD3-short	This paper	N/A
Human HeLa PLVX-HBB-BRWD3-long	This paper	N/A
Human HeLa PLVX-HBB-OGFRL1-short	This paper	N/A
Human HeLa PLVX-HBB-OGFRL1-long	This paper	N/A
Human HeLa PLVX-HBB-m ⁶ A-FUBP1-short	This paper	N/A
Human HeLa PLVX-HBB-m ⁶ A-FUBP1-long	This paper	N/A
Human HeLa PLVX-HBB-m ⁶ A-PAS1-FUBP1-long	This paper	N/A
Human HeLa PLVX-HBB-m ⁶ A-PAS2-FUBP1-long	This paper	N/A
Human HeLa PLVX-HBB-m ⁶ A-PAS3-FUBP1-long	This paper	N/A
Human HeLa PLVX-HBB-m ⁶ A-PAS4-FUBP1-long	This paper	N/A

Oligonucleotides

siRNA-METTL3-sense: CUGCAAGUAUGUUCACUAUGA	This paper	N/A
siRNA-METTL3-antisense: UCAUAGUGAACAUACUUGCAG	This paper	N/A
NXF1 gRNA: TCCAAGGACTATTACAGGG	This paper	N/A

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Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
3' APP DNA linker for GLORI: NNAGATCGGAAGAGCACACGTCTGAA, 5' rApp and 3' NH2 (C6)	This paper	N/A
5' adapter for GLORI: NNNNNNAGATCGGAAGAGCGTCGTGT, 5' Phos and 3' spc3	This paper	N/A
Primers for qPCR, CLONE <i>etc.</i>	This paper	See Table S6
FISH probes	This paper	See Table S6
Recombinant DNA		
pEGFP-C1	Clontech	Cat# 6084-1
HBB-gDNA	This paper	N/A
HBB-cDNA	This paper	N/A
psPAX2	Addgene	Cat# 12260
pMD2.G	Addgene	Cat# 12259
pLVX-TetOne-Puro	Clontech	Cat# 631849
PLVX-HBB-CREB1-short	This paper	N/A
PLVX-HBB-CREB1-long	This paper	N/A
PLVX-HBB-DYRK2-short	This paper	N/A
PLVX-HBB-DYRK2-long	This paper	N/A
PLVX-HBB-BRWD3-short	This paper	N/A
PLVX-HBB-BRWD3-long	This paper	N/A
PLVX-HBB-OGFRL1-short	This paper	N/A
PLVX-HBB-OGFRL1-long	This paper	N/A
PLVX-HBB-ppp1r8-FUBP1-short	This paper	N/A
PLVX-HBB-ppp1r8-FUBP1-long	This paper	N/A
PLVX-HBB-ppp1r8-PAS1-FUBP1-long	This paper	N/A
PLVX-HBB-ppp1r8-PAS2-FUBP1-long	This paper	N/A
PLVX-HBB-ppp1r8-PAS3-FUBP1-long	This paper	N/A
PLVX-HBB-ppp1r8-PAS4-FUBP1-long	This paper	N/A
pxPR001	Addgene	Cat# 51761
NXF1-homology arm	This paper	N/A
PKD-OsTIR1(F74G)	This paper	N/A
Software and algorithms		
fastp v0.23.4	Chen ⁷⁰	https://github.com/OpenGene/fastp
STAR v2.7.10a	Dobin et al. ⁷¹	https://github.com/alexdobin/STAR
SAMTools v1.16.1	Danecek et al. ⁷²	https://github.com/samtools/samtools
cutadapt v4.1	Martin ⁷³	https://github.com/marcelm/cutadapt
NGmerge v0.2	Gaspar ⁷⁴	https://github.com/jsh58/NGmerge
GLORI-tools v1.0.0	Liu et al. ⁴⁶	https://github.com/liucongcas/GLORI-tools

RESOURCE AVAILABILITY**Lead contact**

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Yu Zhou (yu.zhou@whu.edu.cn).

Materials availability

This study did not generate new unique reagents. Materials are available on the request to the [lead contact](#).

Data and code availability

- The raw sequencing data from this study are deposited in the Genome Sequence Archive in National Genomics Data Center, China National Center for Bioinformation / Beijing Institute of Genomics, Chinese Academy of Sciences (HRA006964 under project PRJCA024492) that are publicly accessible at <https://ngdc.cnbc.ac.cn/gsa-human>. The western blots and microscopy images are deposited in Mendeley Data and the corresponding DOI is 10.17632/x5njxrdpm5.2.
- This paper does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS**Cell lines**

HeLa and HEK293T cells, obtained from the Cell Bank of the Chinese Academy of Sciences (Shanghai, China), were grown in Dulbecco's modified Eagle's medium (DMEM, Gibco) supplemented with 10% fetal bovine serum (FBS, Hyclone) and 100 U penicillin/streptomycin (Gibco) at 37°C with 5% CO₂.

METHOD DETAILS**Plasmid construction**

To generate the HBB-gDNA and HBB-cDNA reporters, the fragment with m⁶A sites "GAAGAGGACTTGTGCGGAAACGACGAGAA CAGTTGAAACACAACTTGAACAGCTACGGAAGCTCTGTGCGTAA" was ligated into the pEGFP-C1 vector (6084-1, Clontech) between BglII and Sall. Then, the genomic or cDNA sequence of human β -globin (HBB) was amplified and inserted between AgeI and Bgl II, respectively.

To construct a pair of lentivirus reporters with long or short 3'UTRs (Reporter 1), the genomic sequence of HBB was amplified and ligated into the pLVX-TetOne-Puro Vector (PLVX-HBB) (631849, Clontech), and the terminal exon with long or short 3'UTR was amplified and integrated into the vector in the downstream of HBB through homologous recombination using the pEASY-basic seamless cloning and assembly kit (CU201, TransGen Biotech). For Reporter 2, a fragment from mouse *ppp1r8*¹⁵ was inserted downstream PLVX-HBB (PLVX-HBB-ppp1r8), then long or short 3'UTRs of FUBP1 was inserted. For Reporter 3, different pPAS were synthesized and inserted downstream PLVX-HBB-ppp1r8 together with the extended 3'UTR of FUBP1.

siRNA or DNA transfection and stable cell line generation

To knockdown METTL3, HeLa cells were transfected with the siRNA duplex (Gene Pharma) using RNAi max (13778150, Invitrogen) for 48 hrs (METTL3, sense: CUGCAAGUAUGUUCACUAUGA, antisense: UCAUAGUGAACAUACUUGCAG). The cells were transiently transfected with the indicated plasmids using Lipofectamine 2000 (11668019, Invitrogen).

For lentivirus production, each individual PLVX construct was co-transfected with the package plasmids psPAX2 and pMD2.G into the HEK293T cells. After 12 hrs, the cells were replaced with fresh medium and cultured for another 48 hrs. The lentivirus supernatant was collected and added to HeLa cells. The next day, the stable cells integrated with the plasmid were selected with 1 μ g/ml puromycin (HY-B1743A, MCE) for one week.

For constructing the NXF1-AID cell line, a gRNA targeting the region around the stop codon was designed (TCCAAGGACTA TTACAGGG) and inserted into the CRISPR-CAS9 vector pxPR001. The donor plasmid was constructed in the following order: "left homologous arm - miniAID-3 $\times$ Flag - stop codon - BGH polyA sites - SV promoter - hygromycin resistant gene - right homologous arm". These two plasmids were co-transfected into HeLa cells and selected with hygromycin. Single clone cells integrated with miniAID and Flag tag were kept and a plasmid expressing OstTIR1(F74G) was then integrated into it.

Western blot and antibodies

Western blotting was carried out following the standard protocol. The antibodies below were used: anti-METTL3 (15073-1-AP, Proteintech), anti-Flag (F1804, Sigma), anti-GAPDH (10494-1-AP, Proteintech), anti-m⁶A antibody (202003, Synaptic Systems for LAIC-seq), and (E1610S, NEB for RIP-qPCR).

Reverse transcription, PCR, and quantitative PCR

Total RNA was extracted with Trizol (15596018, Invitrogen) following the manufacturer's protocol, and then reverse transcription was performed according to the manufacturer's instructions (Improli, Promega). PCR was performed using Taq DNA Polymerase according to the manufacturer's protocol (EP0401, Thermo Fisher). The PCR amplicons were run on 2% Agarose gel. The qPCR was carried out with 1:2.5 diluted cDNA, 2 $\times$ SYBR Green qPCR Mix (11201ES03, Yeason) and relevant primers for the amplicons, using BioRad CFX384 Real-Time PCR System. The signal was collected with Bio-Rad CFX Manager version 3.1 and the Ct value was used to calculate the relative expression.

m⁶A LC-MS/MS

LC-MS/MS was performed as previously described.⁷⁵ Total RNA with or without METTL3 knockdown was extracted, followed by polyA⁺ selection and rRNA depletion. 100 ng purified RNA was digested by nuclease P1 (M0660S, NEB), followed by treatment of NH₄HCO₃ and alkaline phosphatase (2250A, Takara). After dilution and centrifugation at 12,000 g for 5 mins, the supernatant was separated by reverse phase ultra-performance liquid chromatography on a C18 column with online mass spectrometry detection using SHIMADZU LC-MS 8050. The nucleotides were quantified and adjusted with the standard curve obtained from pure nucleotide standards running with the same batch of samples. The ratio of m⁶A to A was calculated based on the calibrated concentrations.

Cellular fractionation and RNA-seq

Cellular fractionation was described in our previous study.⁴⁰ Briefly, HeLa cells were first treated with trypsin and collected by centrifugation. The cell pellet was then resuspended in cold cytoplasmic lysis buffer (10 mM Hepes pH 7.5, 0.1% NP-40, 60 mM KCl, 1 mM EDTA, 1 mM DTT, and 0.34 M Sucrose) and incubated on ice for 5 mins, then the cell lysate was centrifuged at 3,500 g for 15 mins at 4°C, and the supernatant was the cytoplasm (CY). The pellet of nuclei was further washed with PBS (1mM EDTA included) and then resuspended in cold glycerol buffer (20 mM Tris-HCl pH 7.5, 75 mM NaCl, 0.5 mM EDTA, 0.85 mM DTT, and 1× cocktail, 50% glycerol), followed by the addition of an equal volume of cold nuclei lysis buffer (10 mM HEPES pH 7.5, 1 mM DTT, 7.5 mM MgCl₂, 0.2 mM EDTA, 0.3 M NaCl, 1 M urea, and 1% NP-40). After vortexing for 2–4 seconds and incubation on ice for 2–3 mins, the lysate was centrifuged at 16,000 g for 2 mins. The supernatant was collected as a nuclear soluble (NP) fraction. The pellet was washed with PBS (1 mM EDTA included) and centrifuged again as the chromatin (CH) fraction.

The RNAs from each cellular fraction were extracted with Trizol LS (10296028, Invitrogen) or Trizol (15596018, Invitrogen) following the manufacturer's instructions. The obtained RNAs were further treated with Turbo DNase I (AM2238, Invitrogen) and purified with RNA Clean and Concentrator 5 (R1014, Zymo research). Then RNA-seq was performed according to the manufacturer's protocol using the commercially available kit (NEBNext® Ultra™ II directional RNA library prep with sample purification beads, E7765S, NEB). Deep sequencing was performed using X-Ten or NovaSeq 6000 platform (Illumina).

m⁶A RIP-RT-qPCR

Total RNA or cytoplasmic RNA was extracted with Trizol (15596018, Invitrogen) following the manufacturer's protocol, then mRNA was purified with oligo(dT) magnetic beads (S1419S, NEB) and treated with Turbo DNase I (AM2238, Invitrogen) followed by concentration with RNA Clean and Concentrator 5 (R1014, Zymo research). m⁶A RIP was carried out using the EpiMark N6-methyladenosine Enrichment kit (E1610S, NEB) according to the manufacturer's recommendations with the following modifications. Briefly, 0.1 fmol of m⁶A positive control and m⁶A negative control spike-in RNA was added to 2 µg purified mRNA. 10% of the mixed sample was saved as the input fraction and the remaining sample was incubated with 1 µl anti-m⁶A antibody conjugated with the protein A/G magnetic beads (88802, Pierce) for 2 hrs at 4°C in reaction buffer (150 mM NaCl, 10 mM Tris-HCl, pH 7.5, 0.1% NP-40). Beads were then washed twice with 200 µL reaction buffer, twice with low salt wash buffer (50 mM NaCl, 10 mM Tris-HCl [pH 7.5], 0.1% NP-40), twice with high salt wash buffer (500 mM NaCl, 10 mM Tris-HCl [pH 7.5], 0.1% NP-40). The m⁶A-modified RNA was extracted with Trizol using the MaXtract High Density (129046, Qiagen) and precipitated by ethanol, and the input RNA and the entire precipitated RNA were subject to RT-qPCR analysis as described above. The m⁶A level represents the value of m⁶A antibody immunoprecipitated (IPed) RNA normalized to the input RNA analyzed by quantitative real-time RT-PCR (RT-qPCR).

m⁶A spike-ins synthesis

The CDS region of mCherry, GFP, Renilla luciferase, Kanamycin, and Ampicillin were amplified and used as templates for *in vitro* transcription according to the protocol of the manufacturer (M0251S, NEB). UTP or UTP/4sUTP (NU-1156S, Jena Bioscience) mixture (10:1) was used to transcribe the UTP and 4sUTP-transcribed m⁶A spike-ins. ATP or ATP/m⁶ATP (M6ATP001-1, synthgene) mixture (20:1) were used to transcribe the m⁶A unmodified and modified transcript, respectively. The modified and unmodified were mixed as the indicated ratio: mCherry (80%), GFP (20%), Renilla luciferase (60%), Kanamycin and Ampicillin (0%) for UTP-transcribed spike-ins, while this ratio of 4sUTP-transcribed spike-ins was half of its UTP counterpart. UTP and 4sUTP transcribed m⁶A spike-ins were further mixed (1:1) and used in the 4sU-m⁶A-LAIC-seq assay.

UTP-transcribed m⁶A spike-ins are used as mimics of steady-state RNA and are *in-vitro* transcribed with UTP and then mixed as indicated ratio for m⁶A modified and unmodified transcripts. 4sUTP-transcribed m⁶A spike-ins are used as mimics of nascent RNA and are *in-vitro* transcribed with 4sUTP/UTP mixture (4sUTP:UTP=1:10) and then mixed as indicated ratio (half the ratio of their 4sUTP-counterparts) for m⁶A modified and unmodified transcripts. These two mixtures are further mixed as the final m⁶A spike-ins mixtures before m⁶A RIP.

4sU-m⁶A-LAIC-seq assay

The 4sU-m⁶A-LAIC-seq strategy was modified from the m⁶A-LAIC-seq.²⁴ HeLa cells were cultured in a medium containing 1 mM 4sU (sigma) for 10 mins or 30mins, then chromatin/nuclear matrix (CH) associated RNA was extracted with Trizol as described above. The polyadenylated RNA was further purified from the obtained RNA above using oligo(dT) magnetic beads (S1419S, NEB) and concentrated by ethanol precipitation. The precipitated RNA was then quantified, mixed with m⁶A spike-ins and alkylated under the

condition [10 mM Iodoacetamide (IAA, sigma), 50 mM NaPO₄, 10 µg polyA⁺ RNA, 50% DMSO (sigma), 50 °C for 20 mins]. After that, 1 µl of 1 M DTT was added to stop the reaction. The IAA-treated RNA was precipitated by ethanol precipitation again and digested with Turbo DNase I (Invitrogen) to deplete the potential residual DNA. At last, the RNA was purified with Agencourt RNA Clean XP (Beckman), quantified using RNA HS Assay Kit (Q32852, Invitrogen), and used for downstream m⁶A RNA immunoprecipitation (RIP) assay.

On the day preceding m⁶A RIP, m⁶A antibody without BSA (202003, Synaptic Systems) was coupled to the Dynabeads using the antibody coupling kit as manufacturer's instructions (14311D, Life Technologies) on a roller at 37 °C overnight. On the following day, 200 µg BSA (Amresco) was added to the coupling system and incubated for another 3 hrs to block the beads for reducing the nonspecific binding, then the prepared m⁶A-Dynabeads were washed and equilibrated with 500 µl of m⁶A binding buffer (50 mM Tris-HCl [pH 7.5], 150 mM NaCl, 1% NP-40, 0.05% EDTA). 2 µg purified polyA⁺ RNA and 0.05 ng m⁶A spike-ins were denatured at 65 °C for 5 mins and immediately set on ice for another 2 mins. After that, the RNA was added to the m⁶A-Dynabeads in 500 µl m⁶A binding buffer and rotated on a roller at room temperature for 2 hrs. The tubes containing the samples were placed on a magnet until the solution became clear. The supernatant was carefully collected and placed on ice, which represents the m⁶A negative fraction. While the beads were washed sequentially twice with 500 µl of m⁶A binding buffer, twice with 500 µl low-salt buffer (0.25× SSPE, 1mM EDTA, 0.05% Tween-20, 37.5 mM NaCl), twice with 500 µl high-salt buffer (0.25× SSPE, 1mM EDTA, 0.05% Tween-20, 137.5 mM NaCl), twice with 500 µl TET buffer (T.E. + 0.05% Tween-20). 125 µl of elution buffer (20 mM DTT, 150 mM NaCl, 50 mM Tris-HCl [pH 7.5], 1mM EDTA, 0.1% SDS) was then added to the washed beads and incubated at 42 °C for 5 mins to elute the m⁶A modified RNA. The tubes were placed on a magnet until the solution become clear and the elute was collected in a fresh tube, which represents the m⁶A positive fraction. The elution step was repeated four times and the elute at each step was combined together.

After m⁶A RIP, 5 µl of 1:100 diluted universal ERCC spike-in (Invitrogen) was added to the supernatant and the elute, then RNA was extracted using acid phenol: chloroform: isopropyl alcohol (25:24:1; Solarbio life sciences) by MaXtract High Density (Qiagen). The upper liquid phase was transferred to a fresh 1.5 ml tube and precipitated by ethanol overnight at -20 °C in the presence of 1 µl 20 mg/ml Glycogen.

On the following day, the sample was centrifuged at 4 °C for 30 mins at 16,000g. The pellet was then washed with 70% ethanol and centrifuged for another 10 mins at 4 °C at 16,000g. The pellet was then left to dry at room temperature before it was resuspended in the desired volume of H₂O. The precipitated RNA (Supernatant and elution) and the input RNA was quantified using RNA HS Assay Kit (Invitrogen) and subjected to RNA-seq library construction using the commercially available kit (NEBNext® Ultra™ II directional RNA library prep with sample purification beads, E7765S, NEB). Deep sequencing was performed using X-Ten or NovaSeq 6000 platform (Illumina).

4sU-GLORI

HeLa cells were cultured in a medium containing 1 mM 4sU (Sigma) for 5 mins, then the chromatin/nuclear matrix (CH) associated RNA was extracted with Trizol as described above. The polyA⁺ (polyA⁺) and -minus (polyA⁻) RNAs were separated using oligo(dT) magnetic beads (S1419S, NEB) and concentrated by ethanol precipitation.

The polyA⁺ RNA and polyA⁻ RNA biotinylation was performed using MTSEA Biotin-XX (M5037, UELandy).⁶⁹ After the removal of the unreacted MTSEA Biotin-XX through phenol:chloroform:isopropyl alcohol extraction, RNA was precipitated by ethanol overnight. Next day, the RNA covalent linked to MTSEA Biotin-XX (nascent RNA) was separated from the RNA without MTSEA Biotin-XX attachment (steady-state RNA) using Dynabeads™ MyOne™ Streptavidin C1 beads (65002, invitrogen). Thus, we got four populations of RNAs: nascent polyA⁻ RNA (npolyA⁻), steady-state polyA⁻ RNA, nascent polyA⁺ RNA (npolyA⁺), steady-state polyA⁺ RNA (spolyA⁺). The npolyA⁻ RNA was further treated with antisense probe and thermostable RNase H (R7090S, beyotime) to remove rRNA.⁷⁶ After that, npolyA⁻, npolyA⁺, and spolyA⁺ RNAs were treated with Turbo DNase I to deplete the potential residual DNA and purified with RNA clean XP beads. These three classes of RNAs were subject to RNA-seq described above or GLORI.

GLORI was performed as the original protocol except for some minor modifications.⁴⁶ RNA was first treated with fragmentation reagents (AM8740, Invitrogen). 200 ng fragmented RNA and 50 pg GLORI m⁶A spike-ins were treated in 40 µl RNA protection buffer (1.32M glyoxal [50649, Sigma], 50% DMSO [0231, Amresco]) at 50 °C for 30 mins, then 10 µl saturated H₃BO₃ solution (B0394, Sigma) was added and reacted at 50 °C for another 30 mins. Then, 50 µl deamination buffer (5 M NaNO₂ solution, 15 µl; 500 mM MES buffer [pH 6], 8 µl; 8.8 M glyoxal solution, 10 µl; nuclease free water, 17 µl) was added, mixed, and reacted at 16 °C for 8 hrs. RNA was purified by ethanol precipitation, dissolved in 50 µl deprotection buffer (formamide, 23.75 µl; TEAA buffer [pH 8.6], 25 µl; nuclease free water, 1.25 µl) and reacted at 95 °C for 10 mins. RNA was precipitated by ethanol and purified using RNA Clean & Concentrator kit.

For library construction, the samples were dephosphorylated (RNA sample, 7 µl; 10× T4 PNK buffer, 1 µl; T4 PNK, 1 µl; SUPERase.in, 1 µl) and ligated with 3' adapter (50% PEG8000, 7 µl; 10× T4 ligase buffer, 1 µl; 3' APP DNA linker [20µM], 1 µl; T4 RNA Ligase 2 truncated KQ, 1 µl). Then, 1 µl 5' deadenylase (NEB, M0331S) and 1 µl RecJf (NEB, M024S) were added into the reaction to remove the excess adapters. RNA was purified using oligo Clean and Concentrator (Zymo Research) and used as the template for reverse transcription (0.1 M DTT, 2 µl; 5× protoscript buffer, 4 µl; RT primer [10 µM], 1 µl; 10 mM dNTPs, 1 µl; protoscript RTase, 1 µl; RNase inhibitors, 1 µl). Then, 2.5 µl ExoSAP-IT was added to digest the excess RT primer. RNA was degraded, and cDNA was purified using oligo Clean and Concentrator and ligated to 5' adapter (50% PEG8000, 10 µl; 10× T4 ligase buffer, 2 µl; 5' adaptor [20 µM], 2 µl; T4

RNA Ligase 1, high concentration, 1.6 μ L, DMSO, 0.8 μ L; ATP [100mM], 0.2 μ L). After that, the products were purified with silane beads and amplified. The PCR products were run on 2% agarose gel, and DNA fragments with sizes from 165–250 base pairs were recovered and subject to sequencing with the Illumina HiSeq X-ten plus platform.

Region-specific GLORI

To quantify the m⁶A levels of individual sites for HBB-cDNA/gDNA reporters or HBB-ppp1r8 Reporter 3, GLORI was performed as described above, and then 3'-specific reverse primers that target the vector region (HBB-cDNA/gDNA) or mouse *ppp1r8* (HBB-ppp1r8 Reporter 3) were used to amplify the products in target region.

RNA FISH

The FISH probes (see Table S6) are prepared using the DIG Northern Starter Kit (Roche, #12039672910). HeLa cells with 80% confluence were washed once with 1x PBS, fixed for 15 mins with 4% paraformaldehyde, washed three times with PBS, permeabilized with 0.1% Triton X-100 for 15 mins, and washed twice with 1x PBS and then once with 2x SSC for 10 mins at room temperature. FISH probes were then added at 1 ng/ μ L into hybridization buffer (2x SSC, 10% Dextran Sulfate, 0.002% BSA, 2 U/ μ L RNase inhibitor, 5 μ g/ μ L yeast tRNA) followed by incubating slides for 16 hrs at 50°C under humid environment. Then, the cells were washed for 15 mins each, twice with wash buffer 1 (2x SSC, 0.1% SDS, 50% formamide, pH=7.2), twice with wash buffer 2 (1x SSC, 50% formamide, pH=7.2). After washing, the slides were blocked using blocking buffer (1% BSA, 8% formamide) for 30 mins, incubated with Digoxin antibody (Invitrogen, #700772) diluted to 1:1000 and SC35 antibody (Sigma, #S4045) diluted to 1:1000. The secondary antibodies were rabbit Alexa-594 and mouse Alexa-488 (Invitrogen) diluted to 1:1000 with blocking buffer (1% BSA, 8% formamide). Between each incubation, the slides were washed three times with wash buffer 3 (2x SSC, 8% formamide) for 10 mins each. Image acquisition was performed on a Leica confocal microscope using a 63x oil immersion objective.

QUANTIFICATIONS AND STATISTICAL ANALYSIS

Sequencing data pre-processing

For the strand-specific, paired-end RNA-seq reads from the Illumina platform, the adapters were removed using fastp (v0.23.4)⁷⁰ with default parameters. Then, the trimmed reads were mapped to the human ribosomal RNA sequences (NR_145819.1, NR_137295.1, NR_137294.1, and NR_023363.1) using the STAR (v2.7.10a) program⁷¹ in local mode to remove the reads from rRNAs. Finally, the remaining unmapped clean reads were mapped to the GRCh38.p12 (Human) genome with the genes annotation (GENCODE v29 for GRCh38[human]) using the STAR program (v2.7.10a)⁷¹ in local mode with ENCODE options. Only the proper-paired, uniquely mapped alignments were kept for further analysis, using the SAMTools (v1.16.1)⁷² with the parameters “-f 2 -F 2308 -F 12 -q 255”.

Gene expression calculation

For a specific gene, only the transcript annotated as “known Canonical” in the UCSC Known Genes annotation was chosen for gene expression calculation. The FPKM (Fragments Per Kilobase of exon model per Million mapped fragments) metric was used to quantify the expression level of each transcript. A pair of reads (read1 and read2) was counted as a fragment for paired-end RNA-seq data.

PolyA sites identification using SLAM PAS-seq data

The PAS-seq data were pre-processed and mapped to the genome in the same manner as described above. The 3' ends of the mapped reads were taken as the polyA site. Reads were flagged as internal priming artifacts and removed when there was an A-rich sequence around their polyA sites. An A-rich sequence was defined as 6 or more consecutive As, or 7 or more As in a 10-nt window in the –10 to +15-nt region around the cleavage site. The reads were further filtered out when no PAS signal was found –45 to –10 nt upstream of their polyA sites.⁷⁷ The PolyA sites within a ± 50 nt window were merged as one polyA site cluster, of which the site with the most supported reads was treated as the representative polyA site, and all reads associated with the cluster were summed as the supporting reads for the representative site.

4sU-m⁶A-LAIC-seq data analysis

T-to-C reads and non-T-to-C reads identification

The high-confident T-to-C conversion events were identified when all the following criteria were satisfied: (1) $L \geq 10$, where L represents the minimum distance between the offset of the nucleotide conversion and the boundary of the mapped read; (2) $Q \geq 30$, where Q represents the sequencing quality of the nucleotide with conversion; (3) When the identified nucleotide with conversion is located in the overlapping area of read1 and read2, it should be recognized as the same nucleotide conversion events; (4) The nucleotide conversion event was not annotated as a SNP in the SNP database.

The nucleotide conversion ratio was calculated as $C_{\text{ref-to-alt}} / C_{\text{ref}}$, where $C_{\text{ref-to-alt}}$ represents the total reads number containing the nucleotide with T-to-C conversion, and C_{ref} represents the total reads number covering the nucleotide. When there is at least one nucleotide identified as a high-confident T-to-C conversion event in read1 and/or read2 for a fragment, the fragment is flagged as T-to-C read, from the newly synthesized RNA. Otherwise, the fragment is flagged as non T-to-C read, from the steady-state RNA.

***m*⁶A level quantification**

The clean reads from *m*⁶A-LAIC-seq were mapped to the ERCC and human genome fusion reference sequences using STAR program (v2.7.10a)⁷¹ in local mode with ENCODE options. Only the proper-paired, uniquely mapped alignments were kept for further analysis, using the SAMTools (v1.16.1)⁷² with the parameters “-f 2 -F 2308 -F 12 -q 255”. The counts of fragments for all ERCC RNAs in the elute (*m*⁶A⁺) and supernatant (*m*⁶A⁻) samples were fitted to a linear equation: $\log_2 C_e = \log_2 C_s + R$, where C_e and C_s represent the mapped fragment count of ERCC in the elute and supernatant, respectively. Only the ERCC RNAs with both C_e and C_s larger than 50 were used to fit the linear model. The *m*⁶A level of each transcript was calculated as $C_e / (C_e + C_s \times 2^R)$, where C_e and C_s represent the mapped fragment counts of the endogenous transcript in the elute and supernatant, respectively. The *m*⁶A level of the newly synthesized and steady-state RNAs were calculated using T-C reads and non-T-C reads, respectively.

Percentage of distal isoform usage index of *m*⁶A⁺ and *m*⁶A⁻ transcripts

Identification of distal polyA sites

We first defined the extended region starting from the terminal exon for each gene. For all non-overlapped genes, we extracted the transcription start site (TSS) and termination site (TTS) of all annotated transcripts from the genome annotation GENCODE Release 29 (GRCh38.p12) for humans and recorded the 5' most TSS and the 3' most TTS for each gene. We extracted the terminal exon of the canonical protein-coding transcript in the UCSC Known Genes and performed an extension of 10 kb or to the TSS of the downstream gene in the same strand. We defined the region from the terminal exon to the end site of the extension as “extended terminal exon”.

For each gene, we identified its putative 3' end in the extended terminal exon based on the 4sU-*m*⁶A-LAIC-seq data. We calculated the mean coverage of the first 100 nt and last 100 nt of the extended terminal exon as C_F and C_L , respectively. We required the genes for downstream analysis to satisfy: 1) $C_F \geq 5$ in both the *m*⁶A⁺ and *m*⁶A⁻ fractions and $C_F \geq 10$ in at least one fraction; 2) $C_L / C_F \leq 0.1$ in both the *m*⁶A⁺ and *m*⁶A⁻ fractions. We divided the extended terminal exon into consecutive bins of size 100 nt from the 5' end, and calculated the mean coverage of each bin. We then scanned bins from the 5' to 3' end until the coverage of the current bin and the next four consecutive bins were all less than 10% of C_F . We then recorded the position 300 nt downstream of the current bin as a putative 3' end. We identified the putative 3' ends using the *m*⁶A⁺ and *m*⁶A⁻ fractions separately, and defined the 3' most site as the putative 3' end of the gene.

We next integrated the polyA sites (PASs) identified above from SLAM PAS-seq data to define the distal PAS (dPAS) and proximal PAS (pPAS) for each gene. For PASs located within the region from the terminal exon to the putative 3' end, the dPAS was identified based on the following criteria: the mean coverage of its 200 nt downstream region was less than 5% of C_F , and the mean coverage of the region from its closest upstream PAS to itself was more than 10% of C_F . We identified the dPAS using the *m*⁶A⁺ and *m*⁶A⁻ fractions separately, and defined the 3' most dPAS as the dPAS of the gene. The other PASs located between the 5' end of the terminal exon and the dPAS of the gene were categorized as pPASs.

Calculation of PDUI and differential PDUI

The mean coverage for each region between consecutive PASs was calculated, and the mean coverage of the interval from the terminal exon to the closest pPAS was defined as C_{Total} . For each PAS, the mean coverage of the interval from itself to the downstream PAS was defined as C_{Down} . For all intervals, the 3' end 200 nt was excluded in calculating its mean coverage, and for intervals shorter than 250 nt, their first 50 nt were used. For a specific pPAS, the PDUI was calculated as C_{Down} / C_{Total} . Fisher's exact test was used to determine the *P*-value of PDUI difference for each pPAS between the *m*⁶A⁺ and *m*⁶A⁻ fractions, followed by the Benjamini-Hochberg procedure to compute the false-discovery rate (FDR). The pPAS with $FDR \leq 0.05$ was considered as differentially used pPAS when the absolute mean difference of PDUIs between the *m*⁶A⁺ and *m*⁶A⁻ fractions was equal or larger than 0.15. When there was more than one such pPAS for a gene, only the most significant one was kept for downstream analysis.

4sU-GLORI data analysis

Data preprocessing

For the GLORI sequencing reads from the Illumina platform, the adapters were removed using the cutadapt program (v4.1)⁷³ with parameters “-m 9 -q 20”. Then, the 3'-overhangs of the reads were removed by NGmerge (v0.2)⁷⁴ in adapter-removal mode with parameters “-p 0.1 -e”. The paired-end reads with overlap were combined into a single read by NGmerge (v0.2)⁷⁴ in stitch mode with default parameters. Next, the PCR duplicates and UMI were removed by fastp (v0.23.4)⁷⁰ with parameters “-dedup -l 25 -trim_front 16 -trim_tail 12” and the UMI information was kept.

***m*⁶A level quantification and comparison**

The clean reads were mapped to the GRCh38.p12 (Human) genome with the genes annotation (GENCODE v29 for GRCh38[human]), and the *m*⁶A levels of individual *m*⁶A-modified sites were computed by GLORI-tools (v1.0.0)⁴⁶ with parameters “-c 15 -C 0 -p 0 -r 0 -p 1.1 -adp 1.1 -s 0.8 -combine -rvs_fac” for each of two replicates of three samples (npolyA⁺, npolyA⁻, and spolyA⁺).

The reliable *m*⁶A sites of each replicate were required to satisfy all the following criteria: (1) AGcov ≥ 30 , where AGcov represents the coverage of A and G; (2) Acov ≥ 5 , where Acov represents the coverage of A; (3) *m*⁶A level ≥ 0.1 ; (4) *p* value ≤ 0.005 ; (5) RNAcov ≥ 5 , where RNAcov represents the coverage of RNA-seq for the corresponding sample; (6) Q ≥ 30 , where Q represents the average base quality of the *m*⁶A site; (7) no continuous 4 nt same base in nearby 9 nt sequence of the *m*⁶A site. The common sites of the reliable *m*⁶A sites in two replicates were counted as the reliable *m*⁶A sites in the corresponding sample. The union of reliable *m*⁶A sites was used to compare two samples.

Statistical analysis

Parametric and non-parametric comparisons were performed in R using the statistical tests described in figure legends. The linear equations were fitted by the least squares method. Fisher's exact test was used to determine the *P*-value, followed by the Benjamini–Hochberg procedure to compute the false-discovery rate (FDR). Data with error bars are represented as mean $\pm$ SD and each point represents the value of one replicate.